Flow Matching Predictive Control with Constraints

Master’s Thesis in Robotics, Cognition, Intelligence

Author: Liming Liu

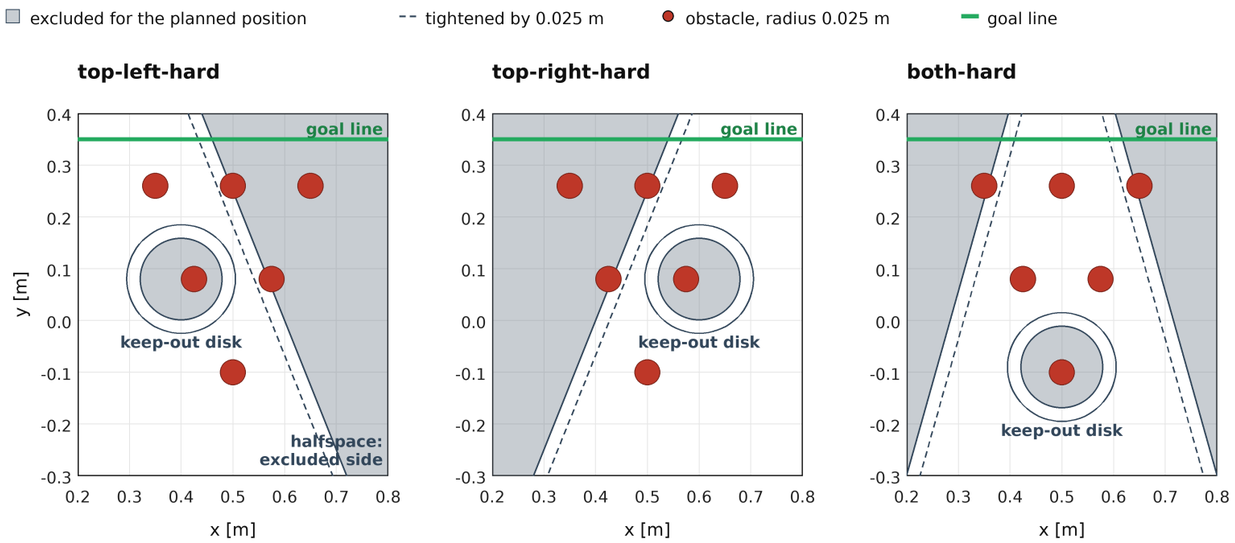
Examiner: Prof. Dr.-Ing. habil. Alois Knoll

Supervisor: Erdi Sayar

Technische Universität München · School of Computation, Information and Technology — Informatics

A generative planner kept safe at inference time, without retraining

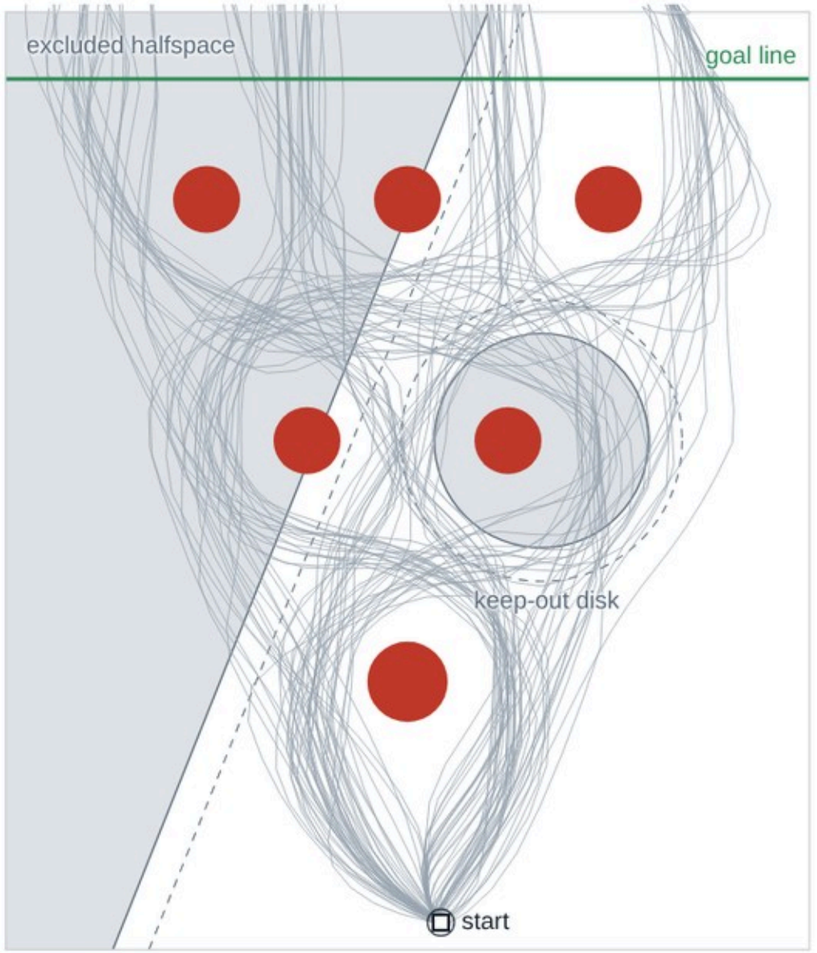
The constraints are declared at deployment and imposed on every plan as it is generated, so when an obstacle is added or moved, the constraint set changes and the trained model stays.



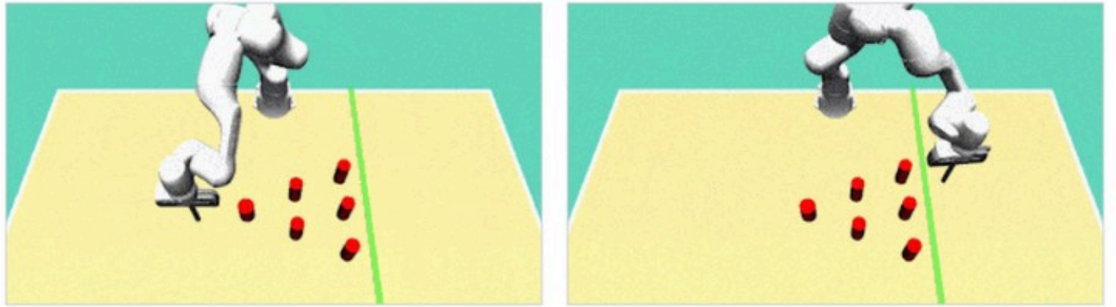
The three constraint geometries of D3IL-avoiding, introduced at test time. Thesis Fig. 5.3

- **The price.** DPCC does this with a diffusion model and a projection at every denoising step, at the price of a long sampling loop run at every control step.
- **The aim.** This thesis asks how to pay less for the same safety, with a flow-matching model in place of the diffusion model.

D3IL-avoiding, the benchmark of DPCC



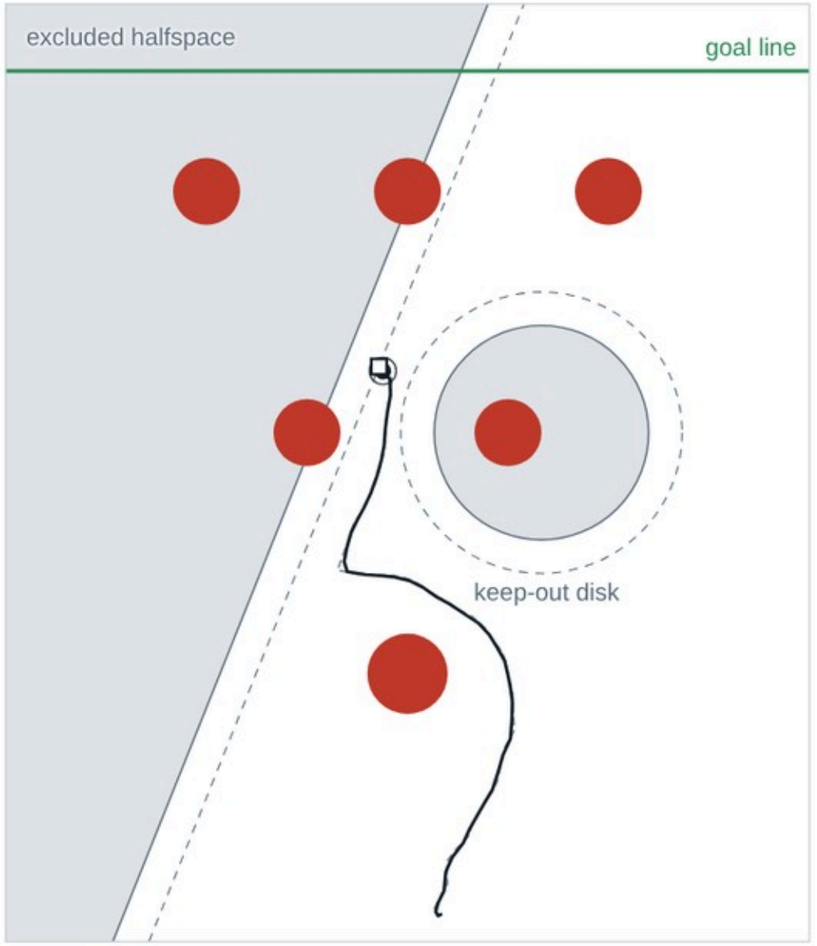
- the 96 recorded human demonstrations (training data)
- obstacle, radius 0.025 m (the lowest 0.03 m)
- goal line
- end effector: measured p (dot), commanded p_{des} (square)
- excluded halfspace and keep-out disk, declared at test time
- the same, tightened by 0.025 m



the scene in MuJoCo: start pose, goal passed · Thesis Fig. 5.2

Illustration, not thesis data · D3IL-avoiding scene, toy planner on its 96 demonstrations

The loop runs once per control step



1 · Observation
$\mathbf{o}_t = (p_{\mathrm{des},t},\ p_t)$

2 · Candidates
$B = 4$ plans, $H = 8$ waypoints, independent noise

3 · Generative model + projection
Diffusion · **FM** · **Analytic-MeanFM** · **CI-MeanFM**
sampling steps k of a flow sampler (here $K = 10$)

4 · Selection
temporal consistency: closest to the previous plan

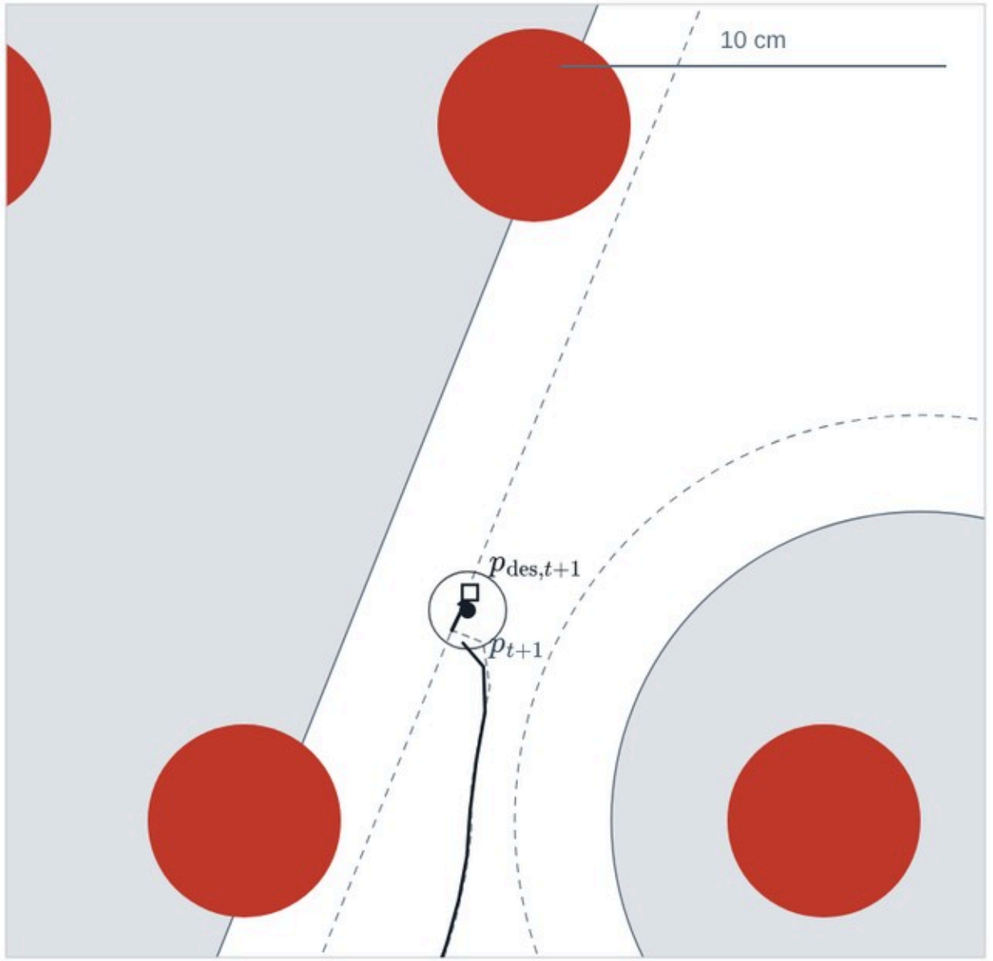
5 · First increment → setpoint
$p_{\mathrm{des},t+1} = p_{\mathrm{des},t} + \Delta p_t$

6 · Tracking controller → plant
differential IK → joint PD → torques → MuJoCo

next control step $t + 1$

Illustration, not thesis data · D3IL-avoiding scene, toy planner on its 96 demonstrations

One control step, block by block



next control step $t+1$

1 · Observation
$\mathbf{o}_t = (p_{\mathrm{des},t},\, p_t)$

2 · Candidates
$B = 4$ plans, $H = 8$ waypoints, independent noise

3 · Generative model + projection
Diffusion · **FM** · **Analytic-MeanFM** · **CI-MeanFM**
sampling steps k of a flow sampler (here $K = 10$) projection active: $k \geq \lfloor (1-\eta)K \rfloor$, $\eta = 0.5$
k = 10 / 10

4 · Selection
temporal consistency: closest to the previous plan

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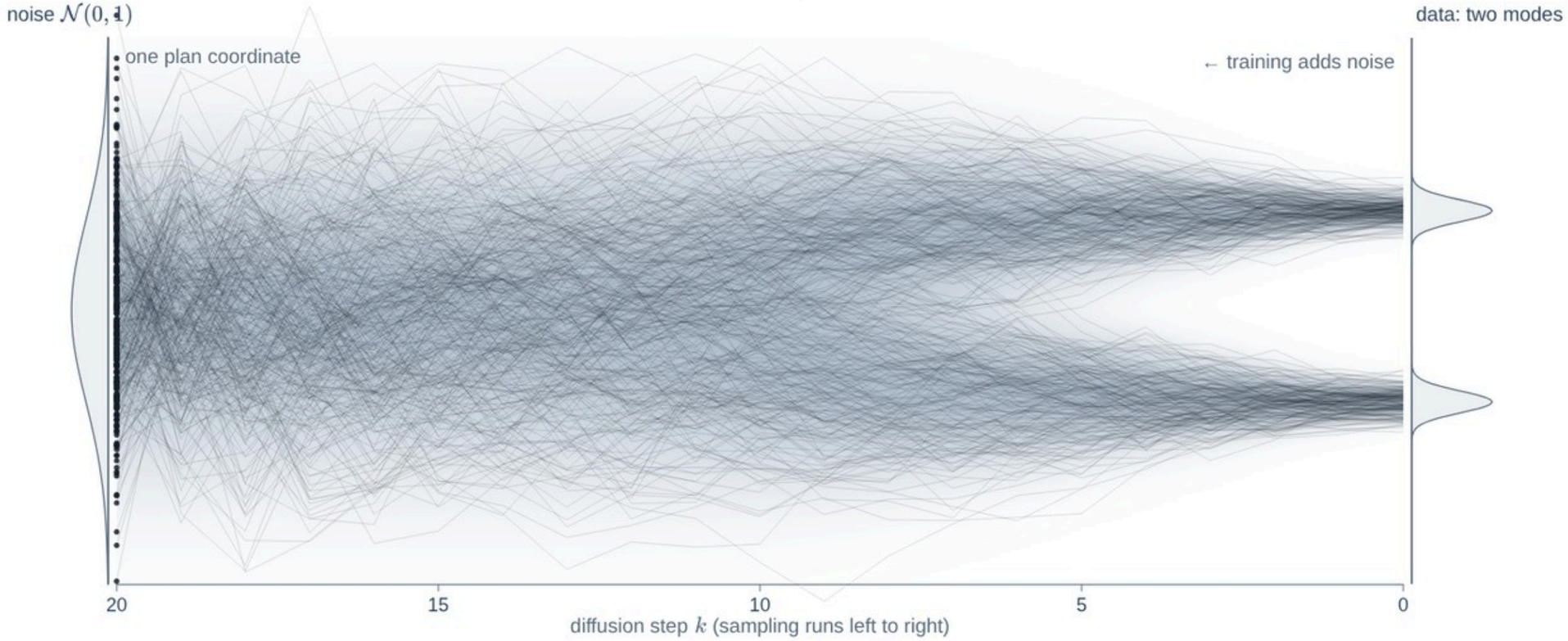
Illustration, not thesis data · D3IL-avoiding scene, toy planner on its 96 demonstrations

A plan is a sample from a learned distribution



Illustration, not thesis data · exact fields of a two-mode toy

Diffusion · train: add noise one step at a time

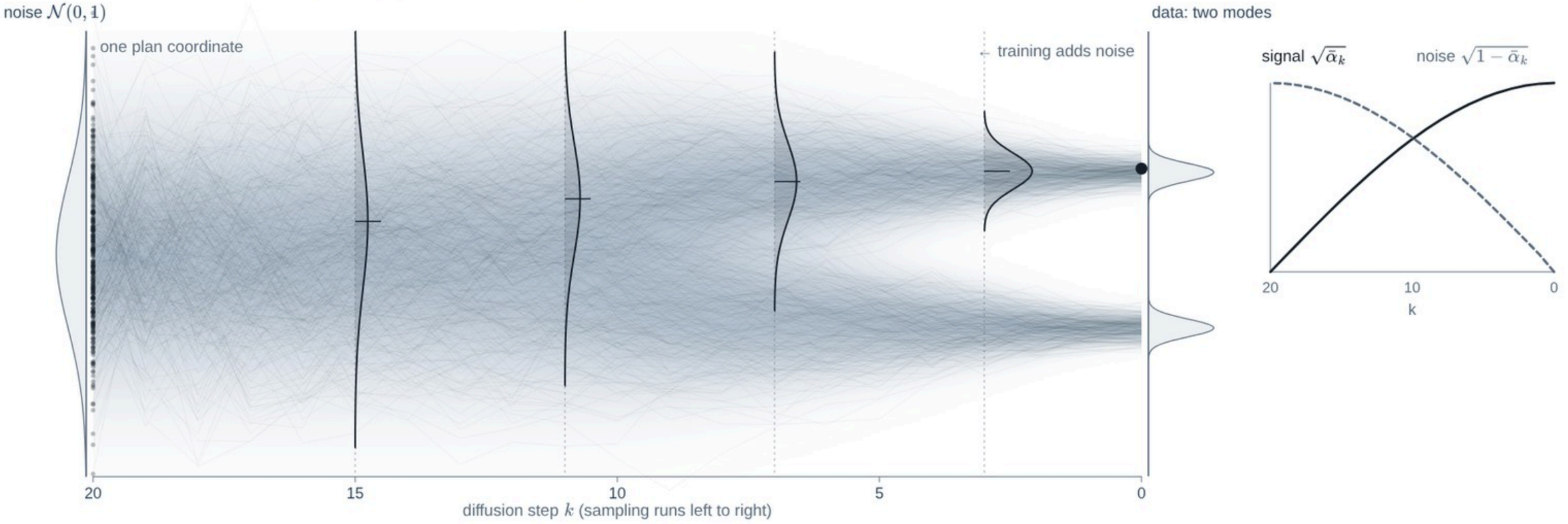


FORWARD STEP

$$\boldsymbol{x}^k = \sqrt{1-\beta_k}\,\boldsymbol{x}^{k-1} + \sqrt{\beta_k}\,\boldsymbol{\epsilon}_k, \qquad \boldsymbol{\epsilon}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad k = 1, \ldots, K$$

Illustration, not thesis data · exact fields of a two-mode toy

Diffusion · any step in one shot

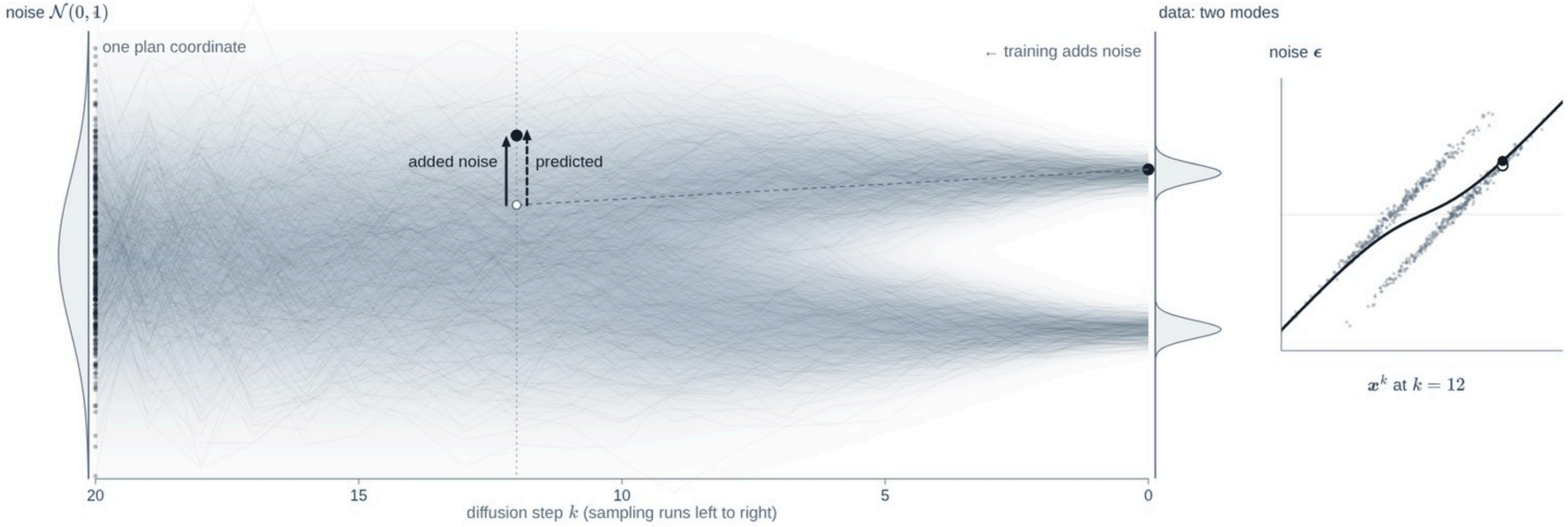


CLOSED FORM OF THE CHAIN

$$\boldsymbol{x}^k = \sqrt{\bar{\alpha}_k}\,\boldsymbol{x}^0 + \sqrt{1-\bar{\alpha}_k}\,\boldsymbol{\epsilon}, \qquad \bar{\alpha}_k = \prod_{i \le k}(1-\beta_i)$$

Illustration, not thesis data · exact fields of a two-mode toy

Diffusion · the network predicts the noise



THE LOSS

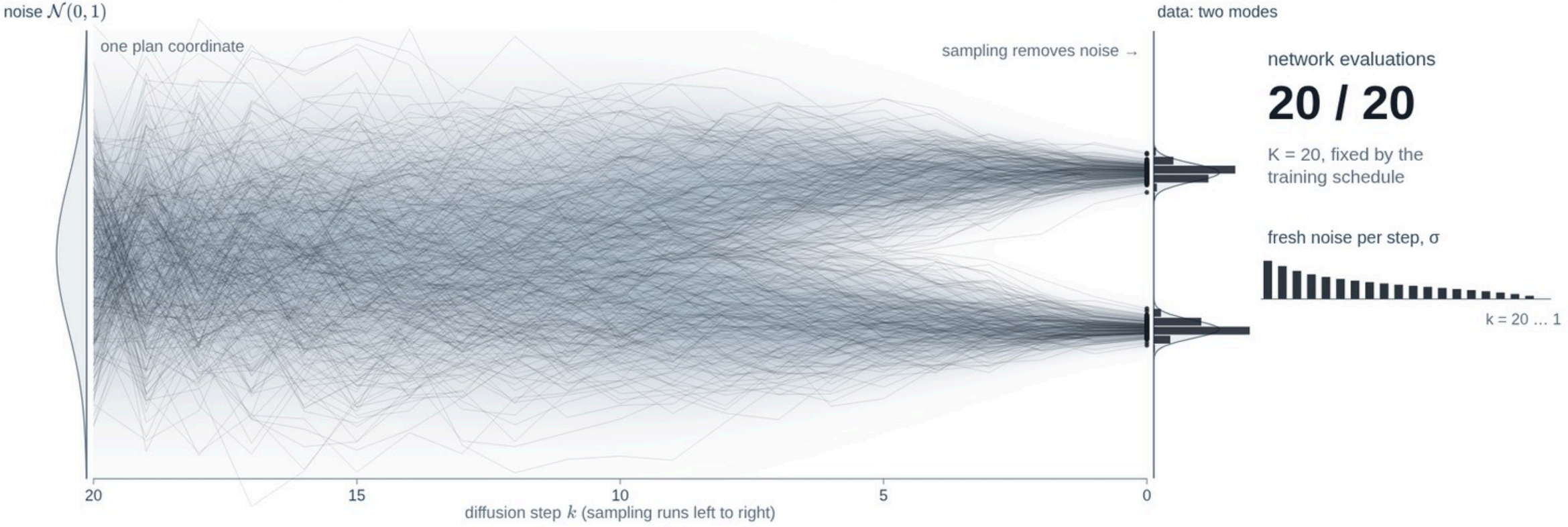
$$\mathcal{L}(\theta) = \mathbb{E}_{k,\, x^0,\, \epsilon} \left\| \epsilon - \epsilon_\theta(x^k, k) \right\|^2$$

ITS MINIMISER: THE MEAN NOISE BEHIND EACH NOISED PLAN

$$\epsilon^\star(x, k) = \mathbb{E}\left[\, \epsilon \mid x^k = x \,\right]$$

Illustration, not thesis data · exact fields of a two-mode toy

Diffusion · sample: remove the noise one step at a time

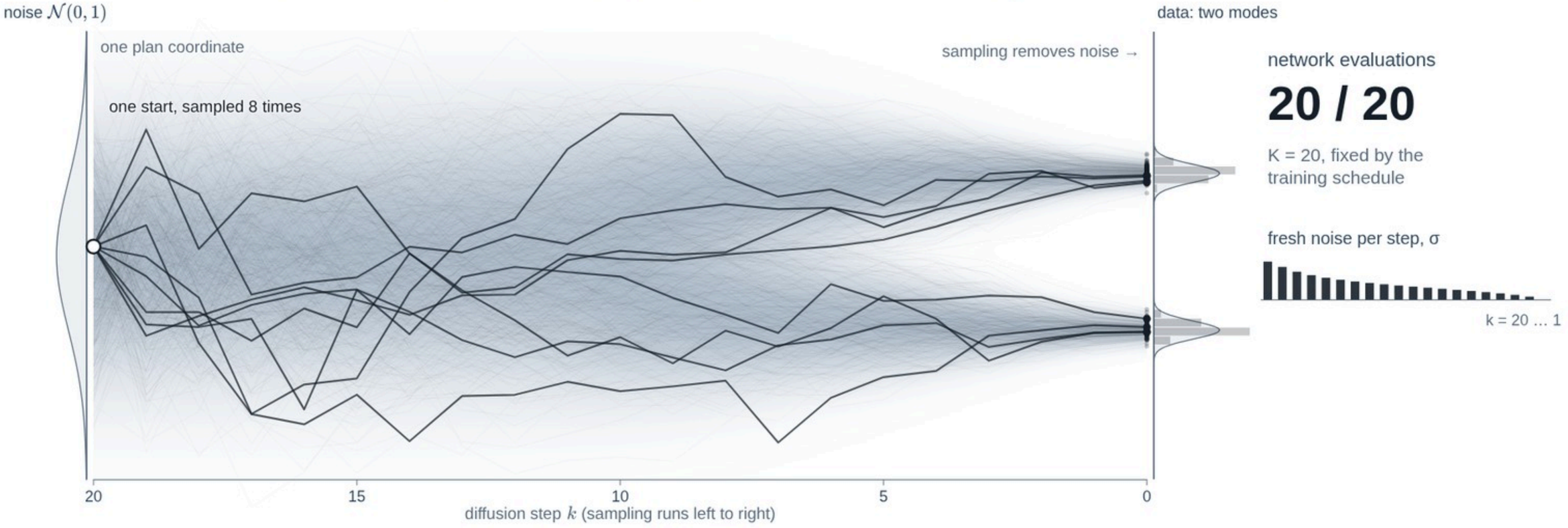


REVERSE STEP (HO ET AL., ALG. 2)

$$\boldsymbol{x}^{k-1} = \frac{1}{\sqrt{\alpha_k}}\left(\boldsymbol{x}^k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\,\boldsymbol{\epsilon}_\theta(\boldsymbol{x}^k, k)\right) + \sigma_k \boldsymbol{z}, \quad \boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Illustration, not thesis data · exact fields of a two-mode toy

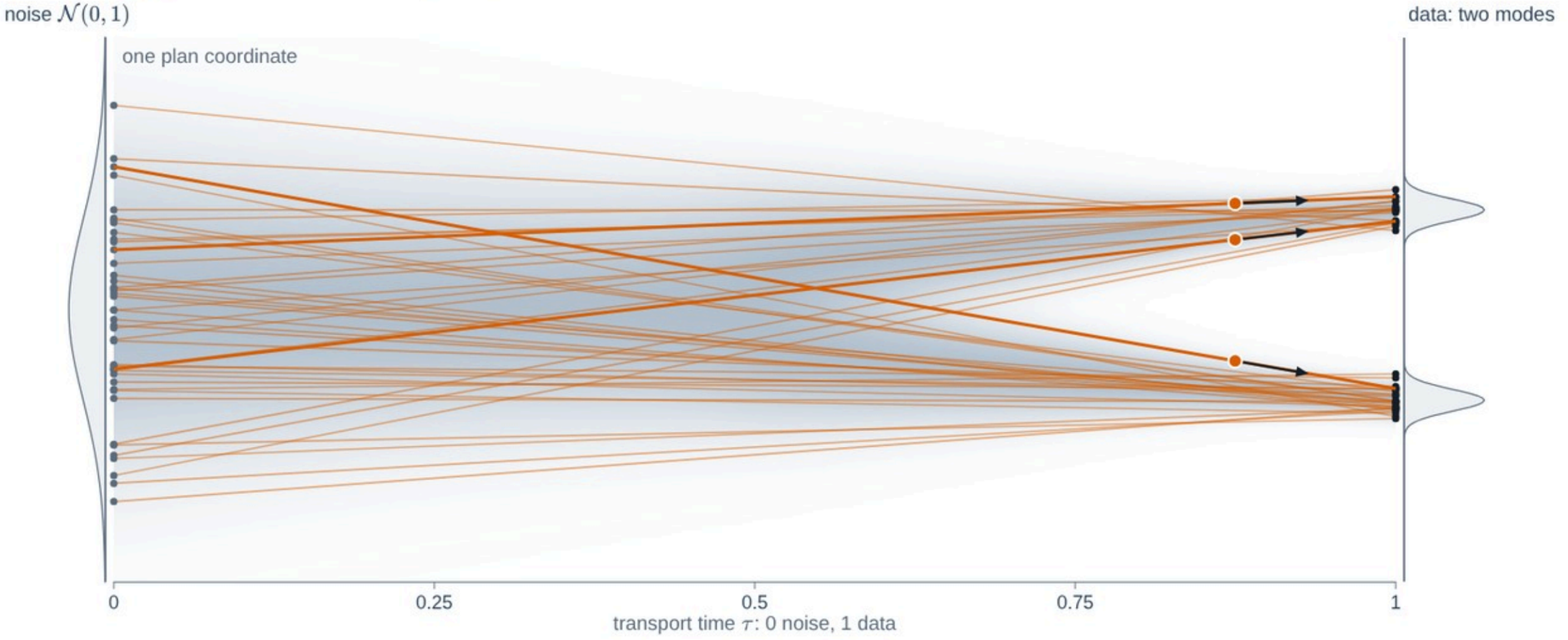
Diffusion · stochastic, with its step count fixed at training



$$\boldsymbol{x}^{k-1} = \frac{1}{\sqrt{\alpha_k}}\left(\boldsymbol{x}^k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\,\boldsymbol{\epsilon}_\theta(\boldsymbol{x}^k, k)\right) + \sigma_k \boldsymbol{z}, \quad \boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Illustration, not thesis data · exact fields of a two-mode toy

FM · train: straight paths from noise to data



THE PATH AND ITS VELOCITY

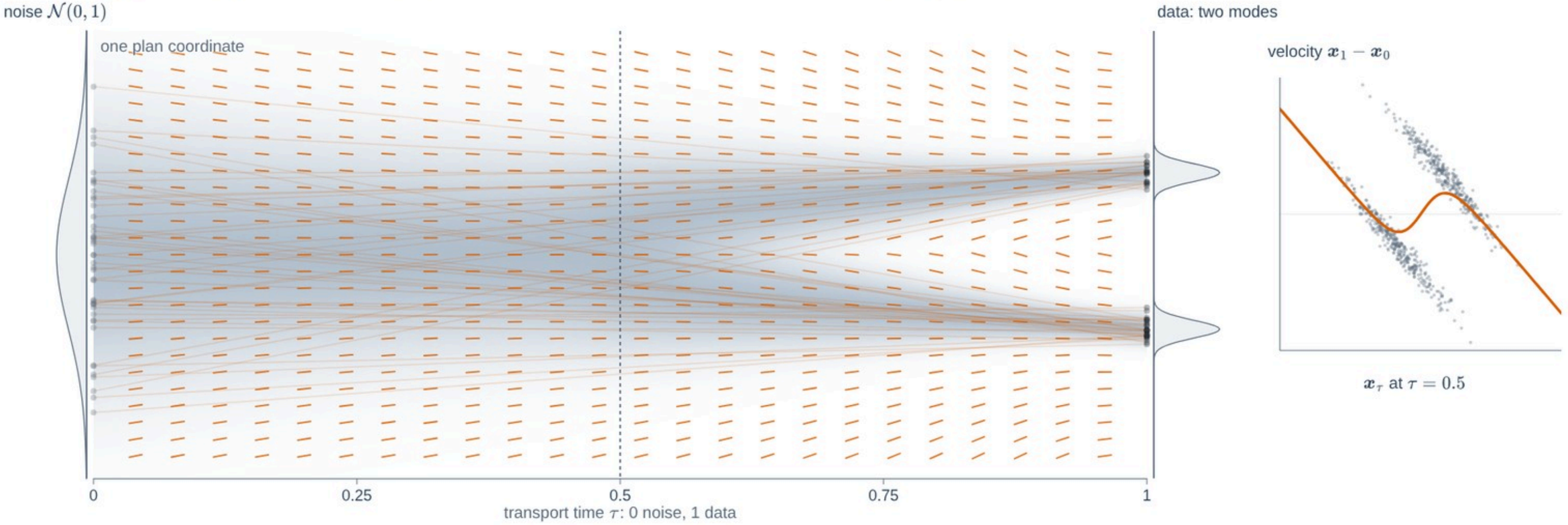
$$\boldsymbol{x}_\tau = (1-\tau)\,\boldsymbol{x}_0 + \tau\,\boldsymbol{x}_1, \qquad \frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = \boldsymbol{x}_1 - \boldsymbol{x}_0$$

THE LOSS

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{\tau,\boldsymbol{x}_0,\boldsymbol{x}_1}\left\| v_\theta(\boldsymbol{x}_\tau,\tau) - (\boldsymbol{x}_1 - \boldsymbol{x}_0)\right\|^2$$

Illustration, not thesis data · exact fields of a two-mode toy

FM · the paths cross; the network learns their average

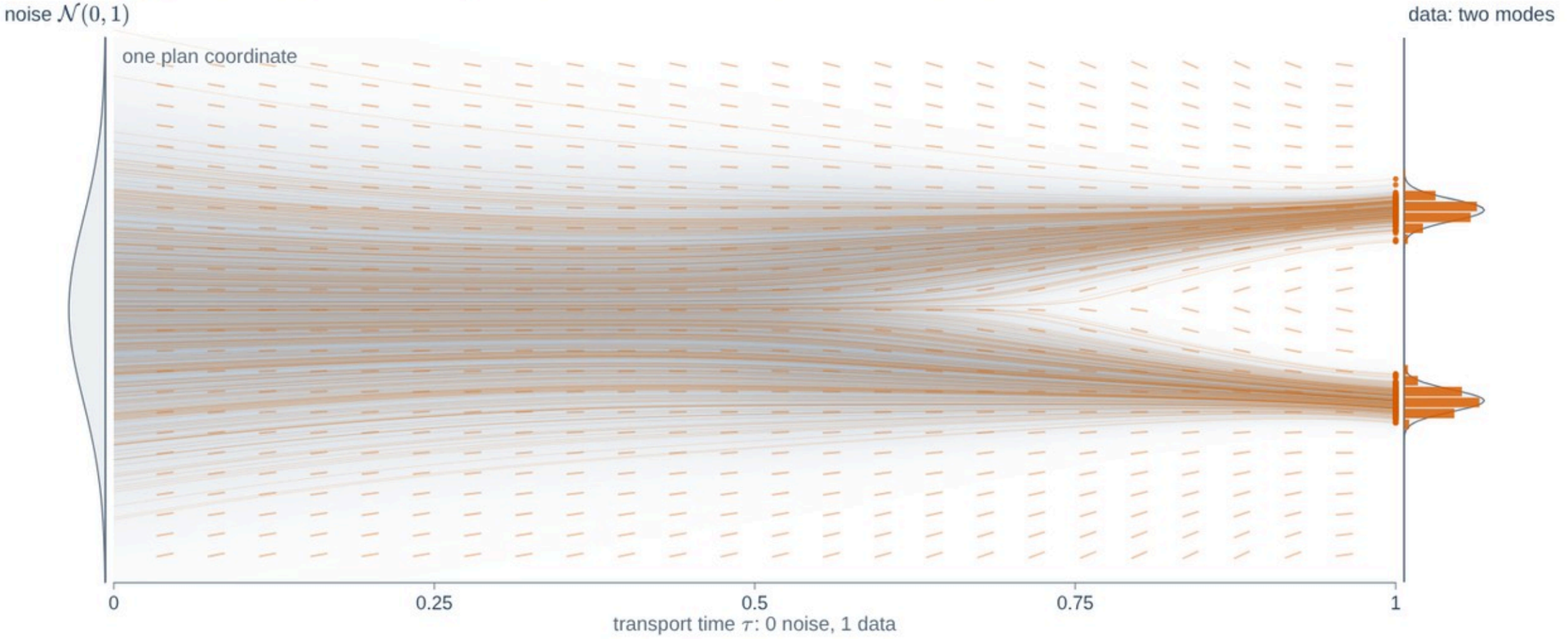


ITS MINIMISER: THE MARGINAL VELOCITY FIELD

$$v(\boldsymbol{x}, \tau) = \mathbb{E}\left[\, \boldsymbol{x}_1 - \boldsymbol{x}_0 \mid \boldsymbol{x}_\tau = \boldsymbol{x} \,\right]$$

Illustration, not thesis data · exact fields of a two-mode toy

FM · sample: integrate the field from noise to data



THE ODE AND ITS SOLUTION

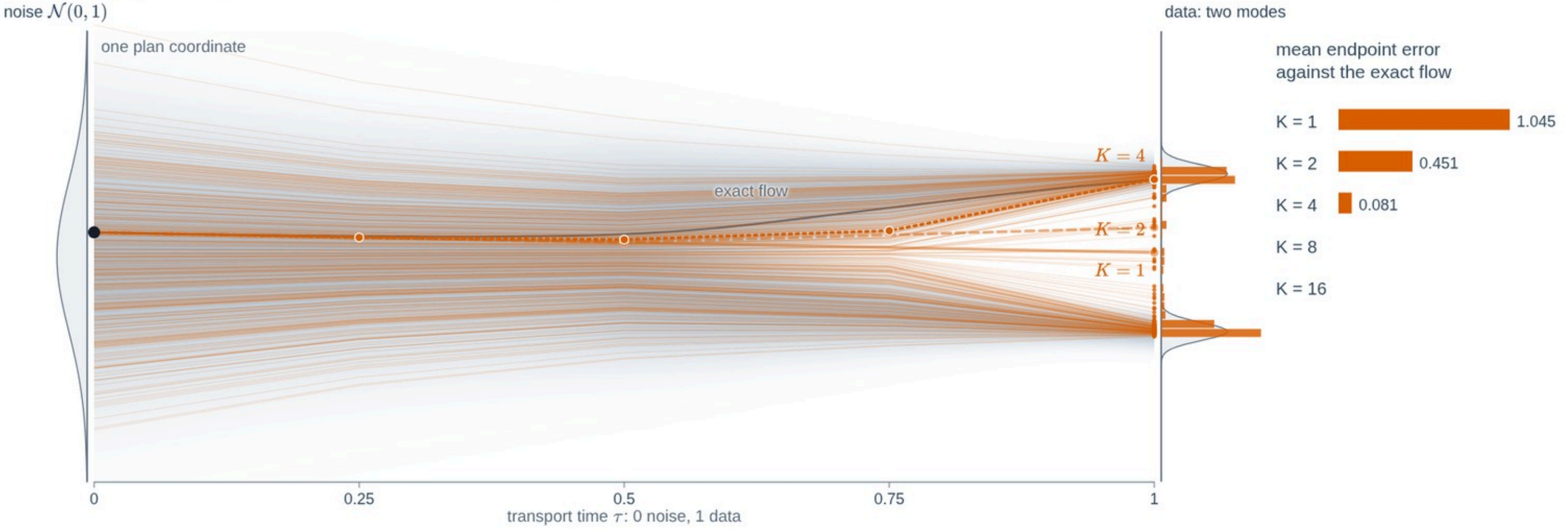
$$\frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = v_\theta(\boldsymbol{x}_\tau, \tau), \quad \boldsymbol{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \implies \boldsymbol{x}_1 = \boldsymbol{x}_0 + \int_0^1 v_\theta(\boldsymbol{x}_\tau, \tau)\,\mathrm{d}\tau$$

CONTINUITY EQUATION

$$\partial_\tau p_\tau + \nabla \cdot \big(p_\tau\, v\big) = 0$$

Illustration, not thesis data · exact fields of a two-mode toy

FM · in practice: K Euler steps

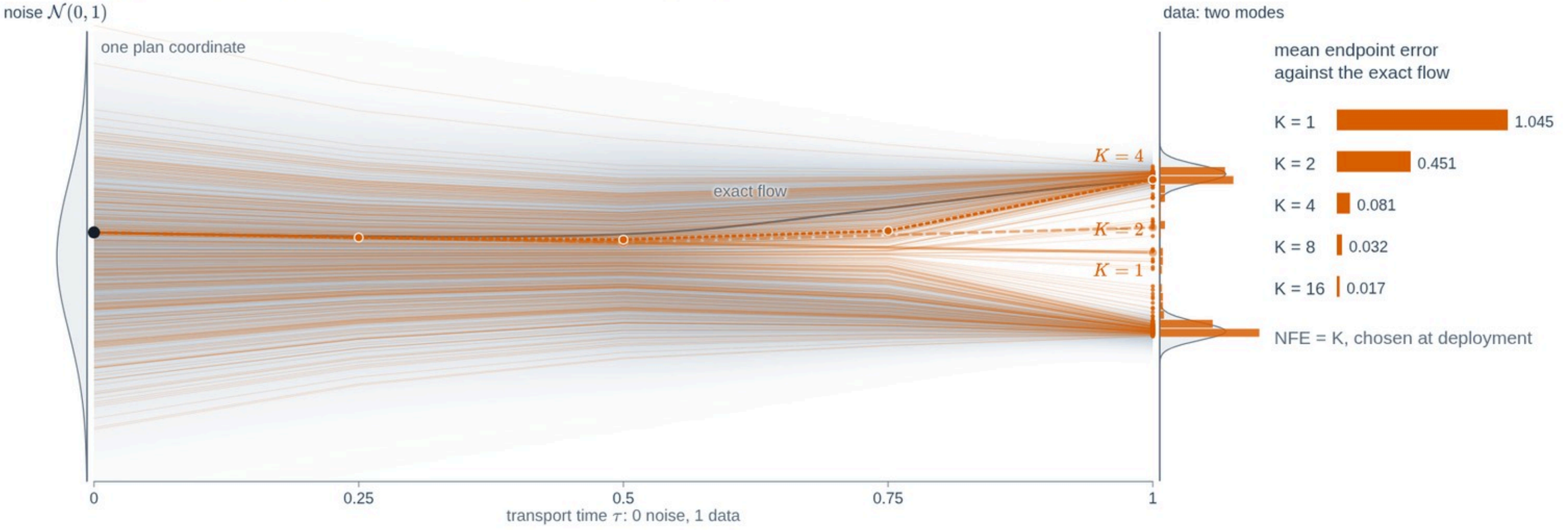


EULER, K STEPS OF $1/K$

$$\boldsymbol{x}_{\tau_{k+1}} = \boldsymbol{x}_{\tau_k} + \frac{1}{K}\, v_\theta(\boldsymbol{x}_{\tau_k}, \tau_k), \qquad \tau_k = \frac{k}{K}$$

Illustration, not thesis data · exact fields of a two-mode toy

FM · the step count is chosen at deployment

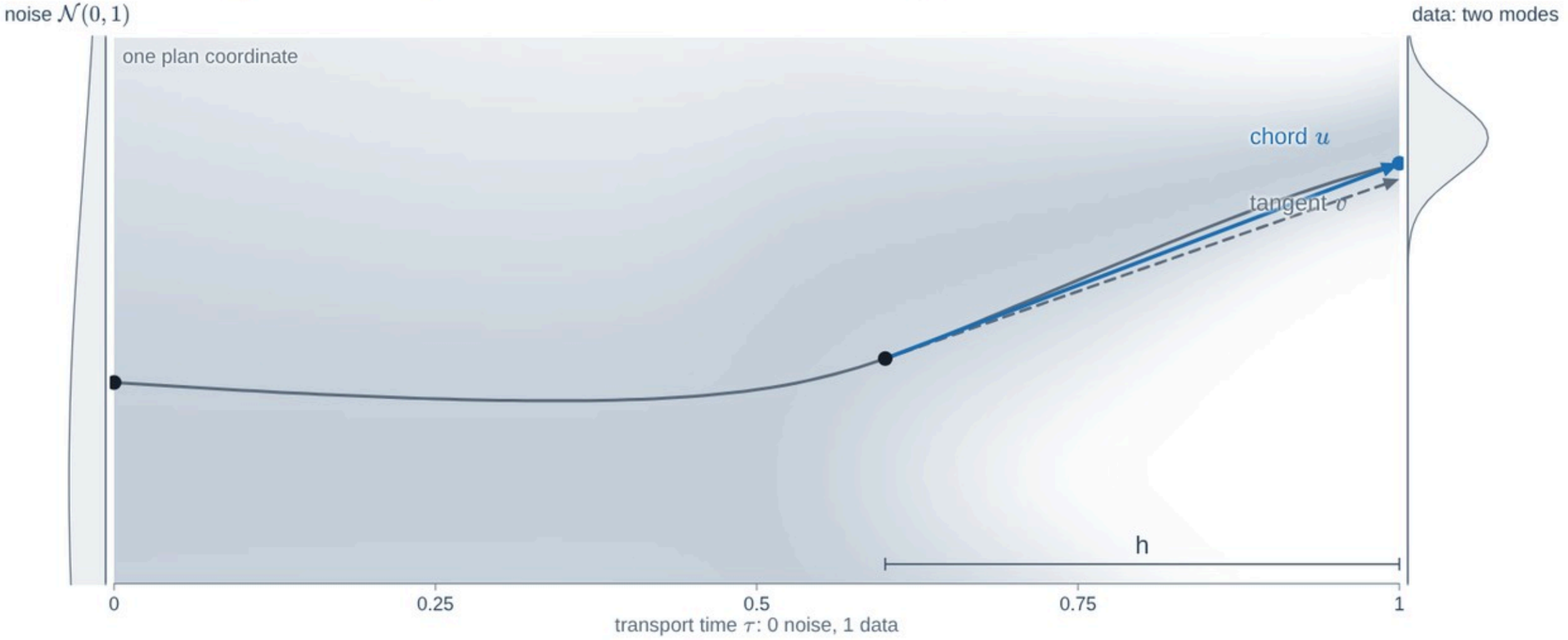


EULER, K STEPS OF $1/K$

$$\boldsymbol{x}_{\tau_{k+1}} = \boldsymbol{x}_{\tau_k} + \frac{1}{K}\, v_\theta(\boldsymbol{x}_{\tau_k}, \tau_k), \qquad \tau_k = \frac{k}{K}$$

Illustration, not thesis data · exact fields of a two-mode toy

Average velocity · the chord, not the tangent

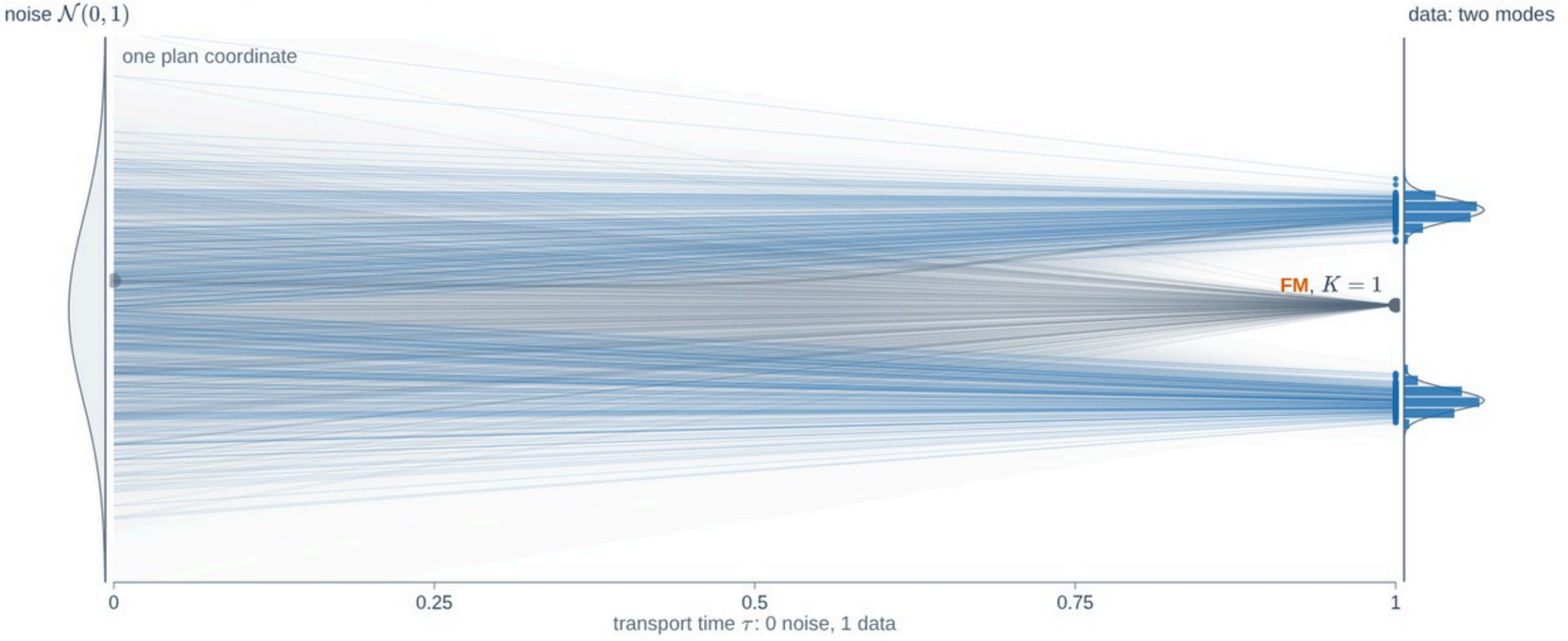


DEFINITION

$$u(\boldsymbol{x}_r, r, h) = \frac{1}{h}\int_r^{r+h} v(\boldsymbol{x}_\varsigma, \varsigma)\,\mathrm{d}\varsigma = \frac{\boldsymbol{x}_{r+h} - \boldsymbol{x}_r}{h}$$

Illustration, not thesis data · exact fields of a two-mode toy

Average velocity · one step covers the interval

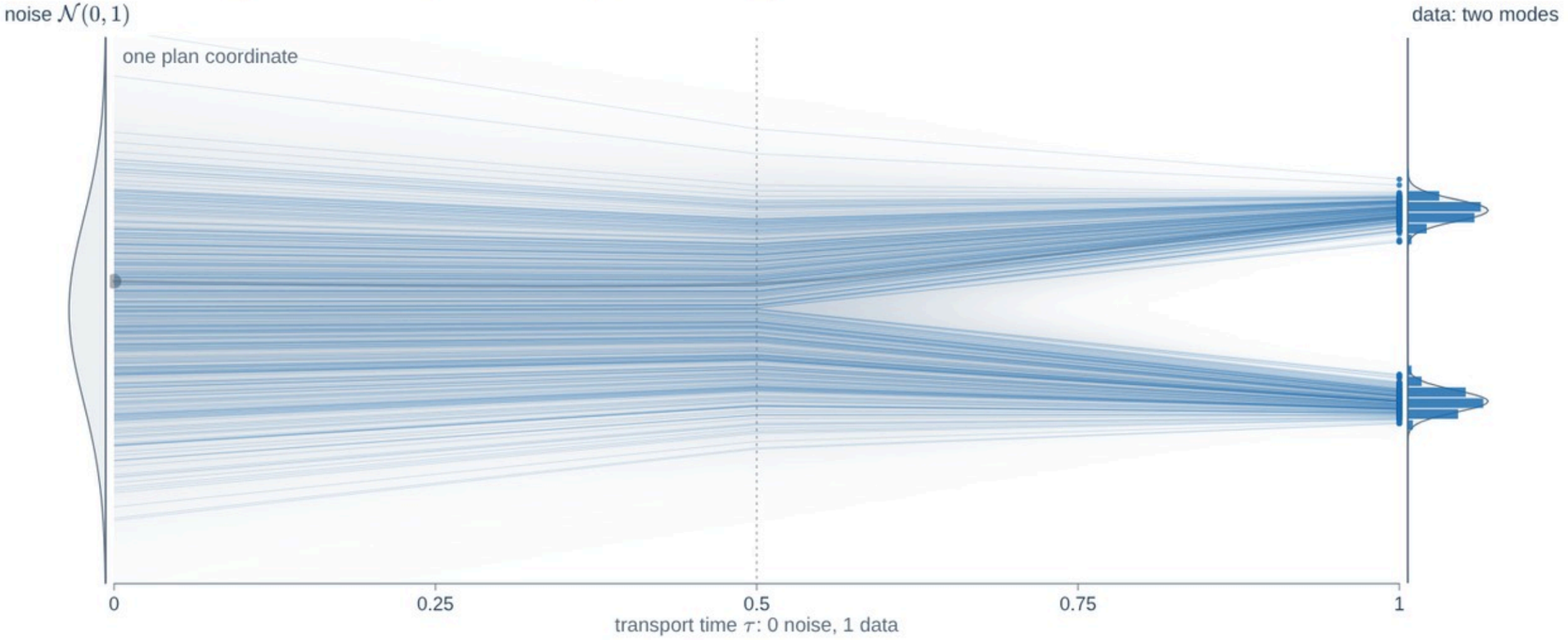


ONE STEP FROM THE NOISE

$$\hat{\boldsymbol{x}}_1 = \boldsymbol{x}_0 + u_\theta(\boldsymbol{x}_0, 0, 1)$$

Illustration, not thesis data · exact fields of a two-mode toy

Average velocity · K steps of length h = 1/K

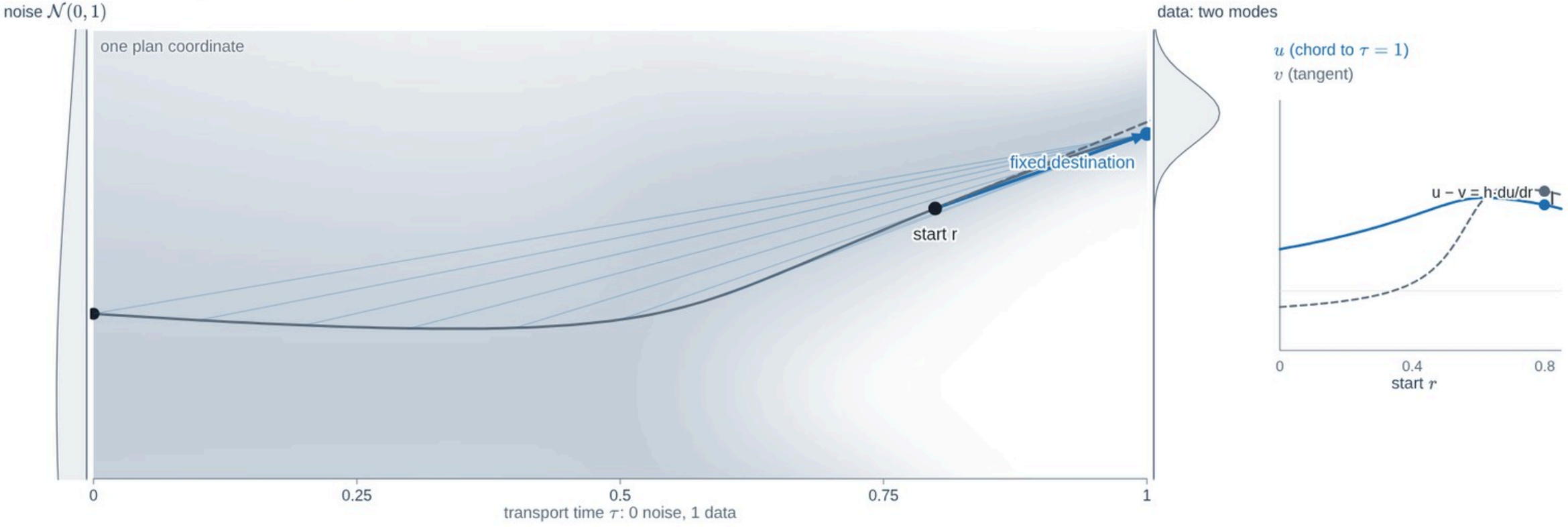


K INTERVAL JUMPS

$$\boldsymbol{x}_{\tau_{k+1}} = \boldsymbol{x}_{\tau_k} + \tfrac{1}{K}\, u_\theta\big(\boldsymbol{x}_{\tau_k}, \tau_k, h{=}\tfrac{1}{K}\big)$$

Illustration, not thesis data · exact fields of a two-mode toy

Average velocity · keep the destination, move the start



THE AVERAGE-VELOCITY IDENTITY

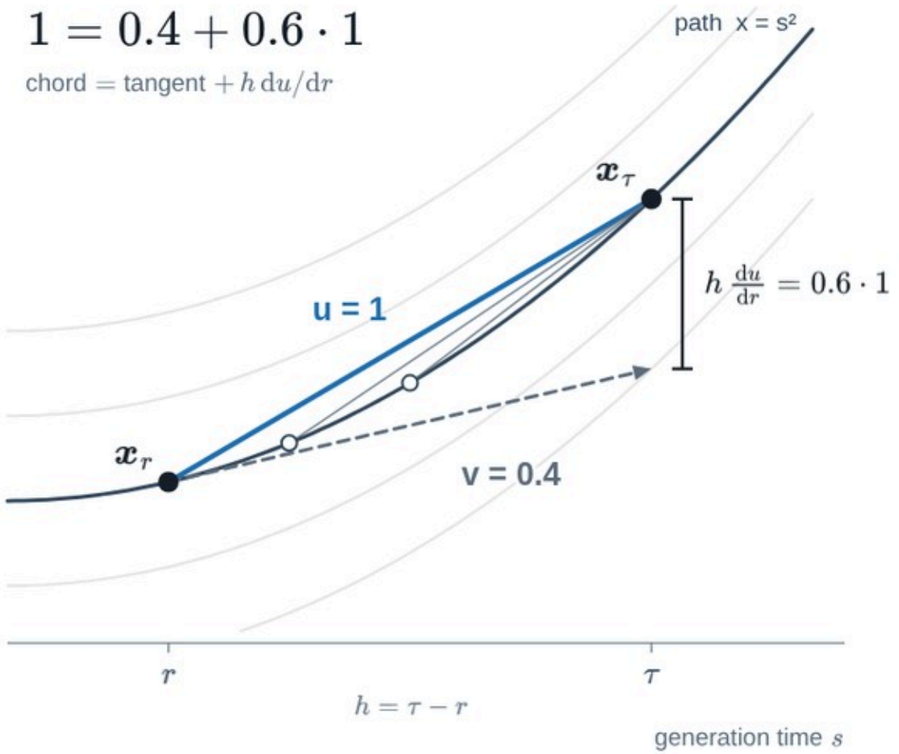
$$u(\boldsymbol{x}_r, r, h) = v(\boldsymbol{x}_r, r) + h\,\frac{\mathrm{d}}{\mathrm{d}r}\,u(\boldsymbol{x}_r, r, \tau - r), \qquad h = \tau - r$$

Illustration, not thesis data · exact fields of a two-mode toy

Analytic-MeanFM · the target from the exact derivative

$$1 = 0.4 + 0.6 \cdot 1$$

chord = tangent $+ h\,\mathrm{d}u/\mathrm{d}r$



THE DERIVATIVE MOVES ALL THREE INPUTS AT ONCE

$$\frac{\mathrm{d}u}{\mathrm{d}r} = (\partial_{\boldsymbol{x}} u)\,\boldsymbol{v} + \partial_r u - \partial_h u = \mathrm{JVP}\big(u;\,(\boldsymbol{x}_r, r, h),\,(\boldsymbol{v}, +1, -1)\big)$$

tangent $(\boldsymbol{v}, +1, -1)$

Jacobian row $(\partial_{\boldsymbol{x}} u,\ \partial_r u,\ \partial_h u)$

0	2	1

$\cdot$

0.4
+1
−1

$= 0 \cdot 0.4 + 2 \cdot 1 + 1 \cdot (-1) = 1$

on this toy $u = 2r + h$: $\partial_{\boldsymbol{x}} u = 0,\ \partial_r u = 2,\ \partial_h u = 1$

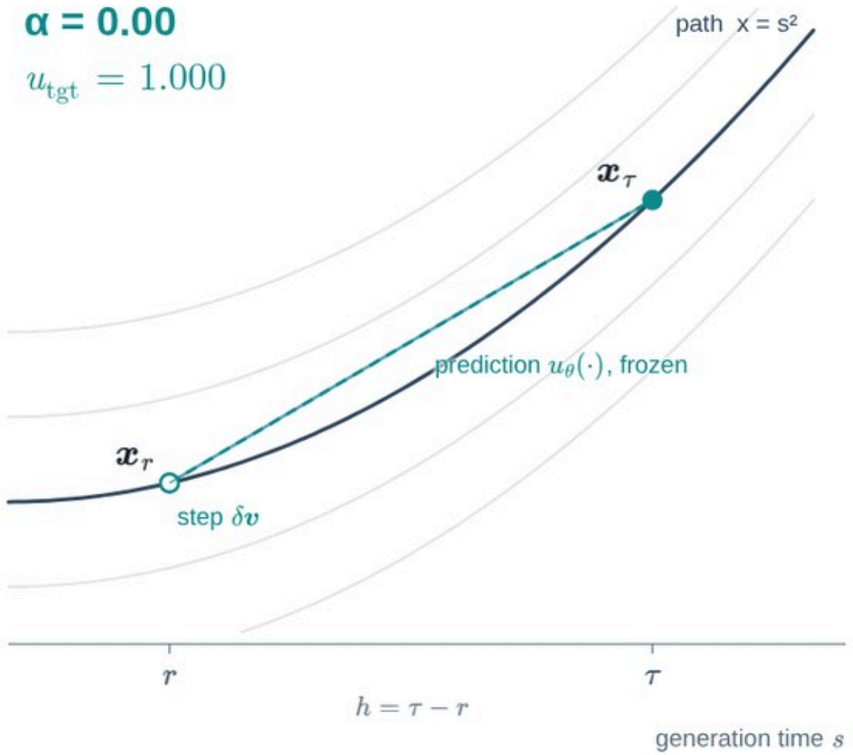
THE ANALYTIC TARGET, FROZEN BY STOP-GRADIENT

$$u_{\mathrm{tgt}} = \mathrm{sg}\left[\,\boldsymbol{v} + h\,\mathrm{JVP}\big(u_\theta;\,(\boldsymbol{x}_r, r, h),\,(\boldsymbol{v}, +1, -1)\big)\right], \qquad \boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$$

$u_{\mathrm{tgt}} = 0.4 + 0.6 \cdot 1 = 1$, the chord: on the exact field the target is exact

Illustration, not thesis data · exact fields of a two-mode toy

CI-MeanFM · a finite step plus the network's own prediction



THE CONSISTENCY-INTERPOLATED TARGET

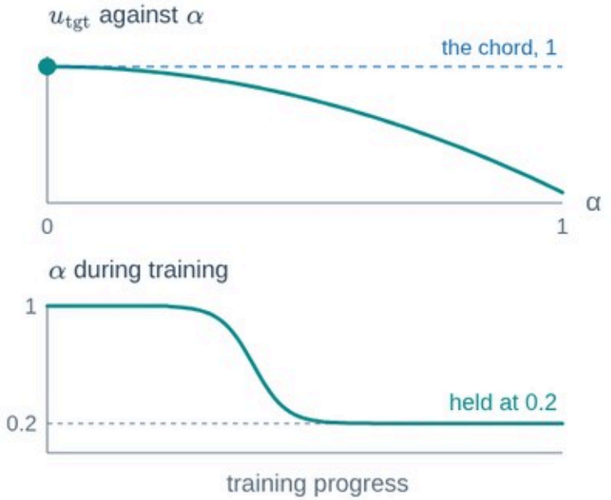
$$h\,u_{\mathrm{tgt}} = \delta\,\boldsymbol{v} + (h-\delta)\ \mathrm{sg}\left(u_\theta(\boldsymbol{x}_r + \delta\boldsymbol{v},\ r+\delta,\ h-\delta)\right), \qquad \delta = \alpha h$$

$\alpha = 1$: $u_{\mathrm{tgt}} = 0.4$

$\alpha = 0.5$: $u_{\mathrm{tgt}} = 0.85$

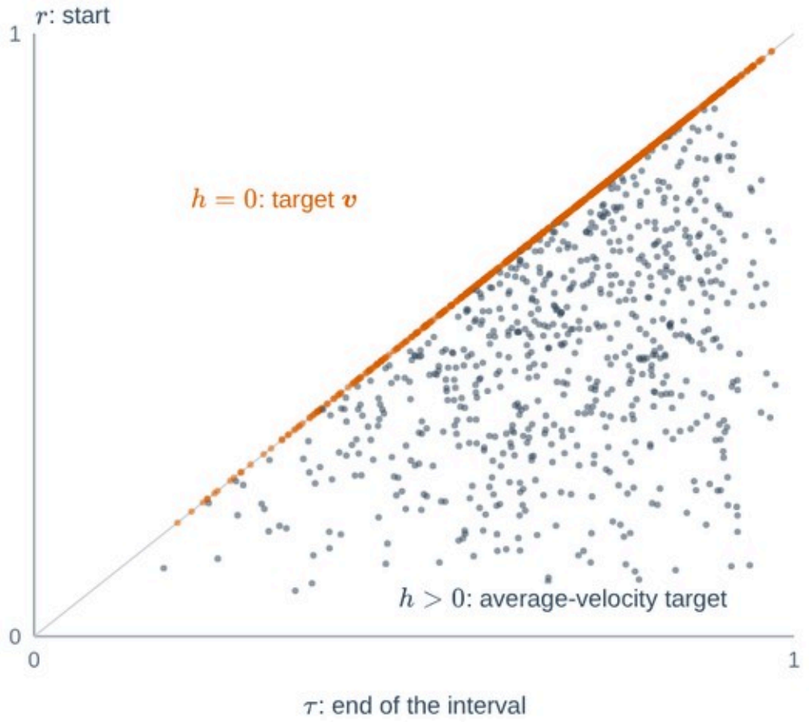
$\alpha = 0.2$: $u_{\mathrm{tgt}} = 0.976$

$\alpha \to 0$: $u_{\mathrm{tgt}} \to 1$



Illustration, not thesis data · exact fields of a two-mode toy

Both · one average-velocity network, half of its samples at h = 0



ZERO-INTERVAL SAMPLES: THE FM TARGET

$$h = 0: \qquad u_{\text{tgt}} = \boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$$

temporal U-Net, shared

- average head $u_\theta(\boldsymbol{x}_r, r, h)$: used by the sampler
- auxiliary head v_θ: regresses $\boldsymbol{v}$ on every sample

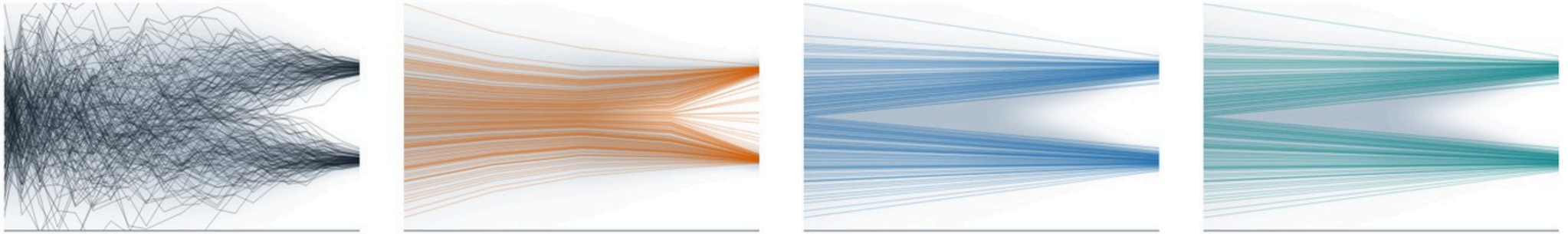
Shared by both average-velocity models: the network and its heads, the $h = 0$ samples, the sampler.

Illustration, not thesis data · exact fields of a two-mode toy

Four generative models inside one constrained controller

	LEARNS	STEPS K	SAMPLING
Diffusion (DPCC)	the noise ϵ	fixed at training ($K = 20$)	adds noise
FM	the velocity v	chosen at deployment	deterministic
Analytic-MeanFM	the average velocity u; target from the exact JVP	chosen at deployment	deterministic
CI-MeanFM	the average velocity u; target from a finite step, α	chosen at deployment	deterministic

Diffusion, $K = 20$ **FM**, Euler $K = 4$ **Analytic-MeanFM**, $K = 1$ **CI-MeanFM**, $K = 1$



the toy with exact fields: both average-velocity models sample the same u; they differ in training

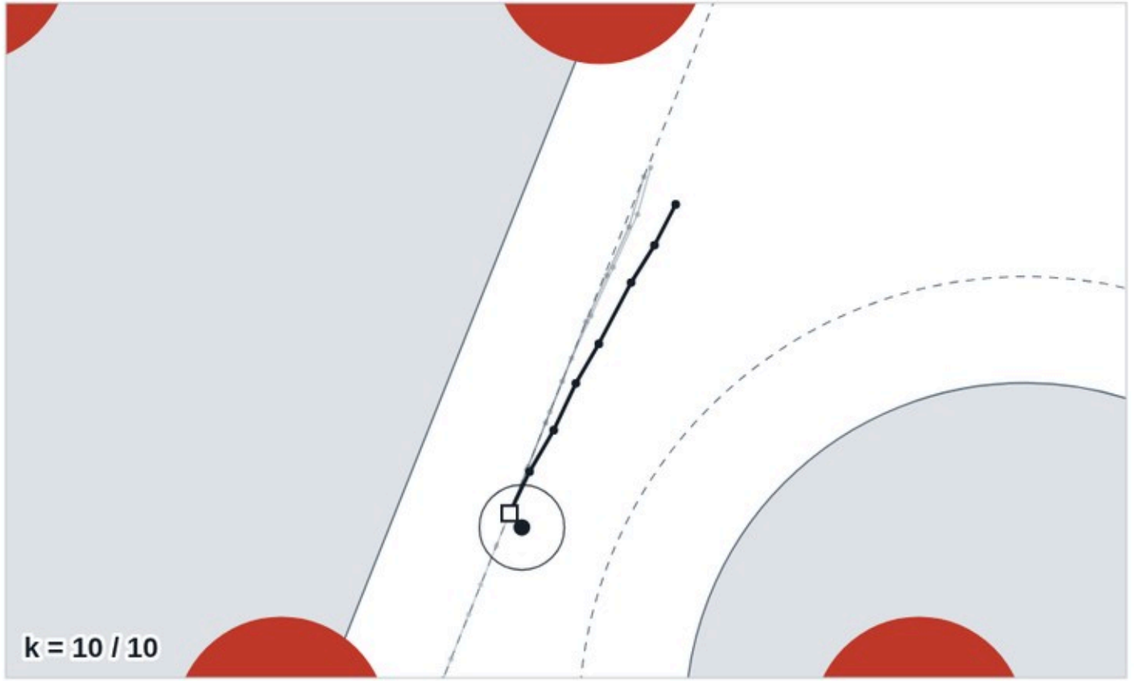
Illustration, not thesis data · exact fields of a two-mode toy

Two ways to put the projection into sampling

Per-step projection (DPCC): project the current iterate

dashed: Euler step · solid: projected

$$\mathrm{NFE} = K = 10$$

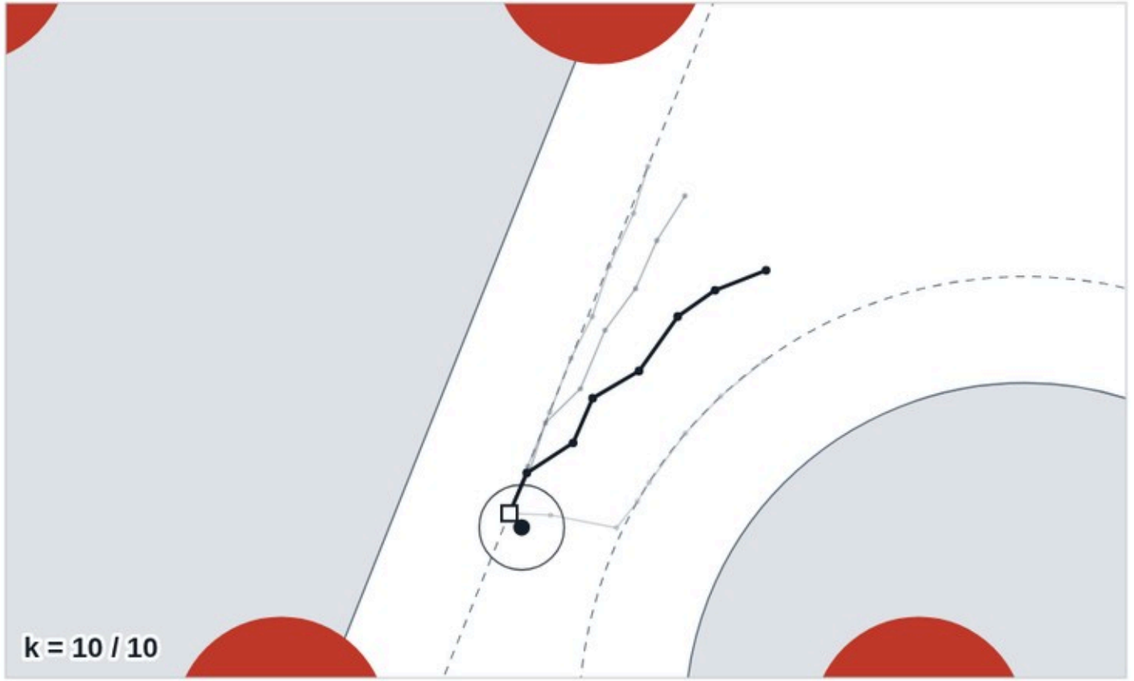


$$\bar{\mathbf{x}}_{k+1} = \mathbf{x}_{\tau_k} + \Delta\tau\, v_\theta(\mathbf{x}_{\tau_k}, \tau_k), \qquad \mathbf{x}_{\tau_{k+1}} = \Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_{k+1})$$

Endpoint projection (HardFlow): project the predicted final plan

dashed: predicted final plan · thin: its projection · solid: the iterate

$$\mathrm{NFE} = K + n_{\mathrm{guide}} = 14$$

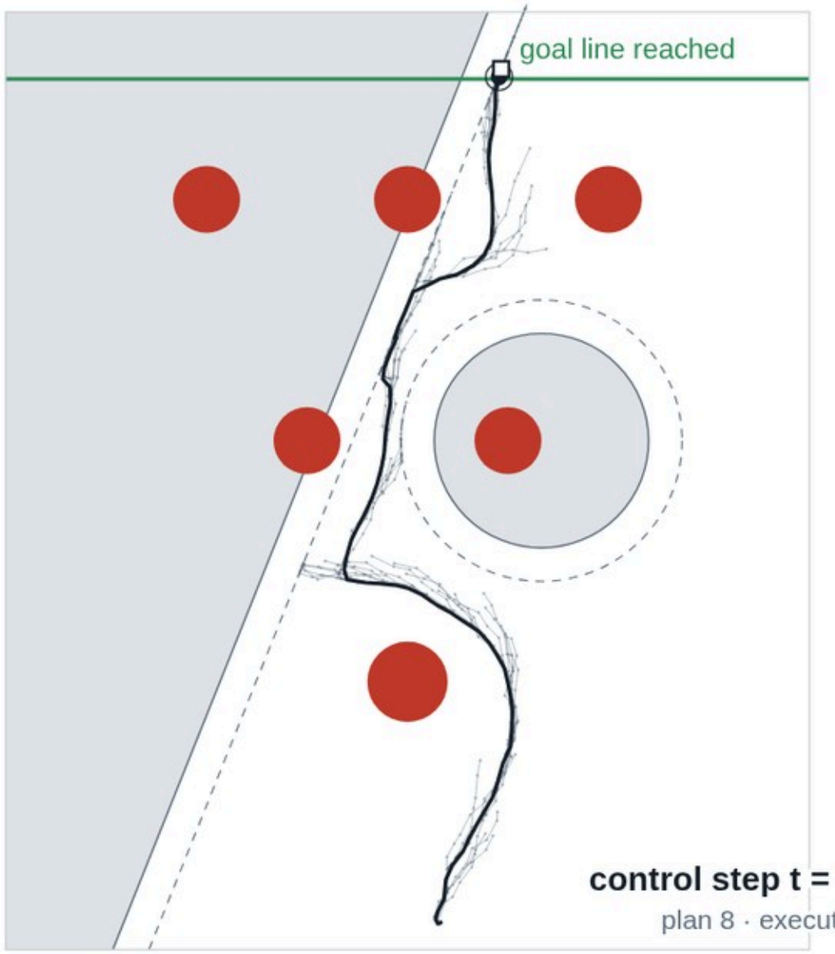


$$\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_{k+1} + (1 - \tau_{k+1})\, v_\theta(\bar{\mathbf{x}}_{k+1}, \tau_{k+1}), \qquad \hat{\mathbf{x}}_1^\star = \Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_1)$$

$$\mathbf{x}_{\tau_{k+1}} = \bar{\mathbf{x}}_{k+1} + \tau_{k+1}\left(\hat{\mathbf{x}}_1^\star - \bar{\mathbf{x}}_1\right)$$

Illustration, not thesis data · D3IL-avoiding scene, toy planner on its 96 demonstrations

Receding horizon: plan eight steps, execute the first, plan again



1 · Observation
$\mathbf{o}_t = (p_{\mathrm{des},t},\ p_t)$

2 · Candidates
$B = 4$ plans, $H = 8$ waypoints, independent noise

3 · Generative model + projection
Diffusion · **FM** · **Analytic-MeanFM** · **CI-MeanFM**
sampling steps k of a flow sampler (here $K = 10$) projection active: $k \geq \lfloor (1-\eta)K \rfloor$, $\eta = 0.5$
k = 10 / 10

4 · Selection
temporal consistency: closest to the previous plan

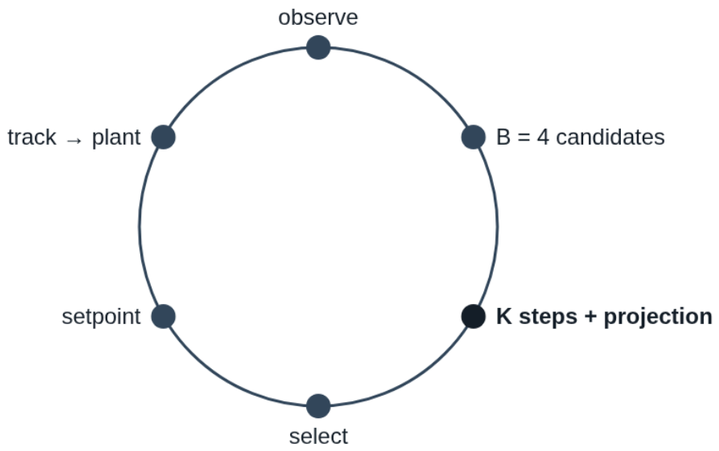
5 · First increment → setpoint
$p_{\mathrm{des},t+1} = p_{\mathrm{des},t} + \Delta p_t$

6 · Tracking controller → plant
differential IK → joint PD → torques → MuJoCo

next control step $t+1$

Illustration, not thesis data · D3IL-avoiding scene, toy planner on its 96 demonstrations

Research questions

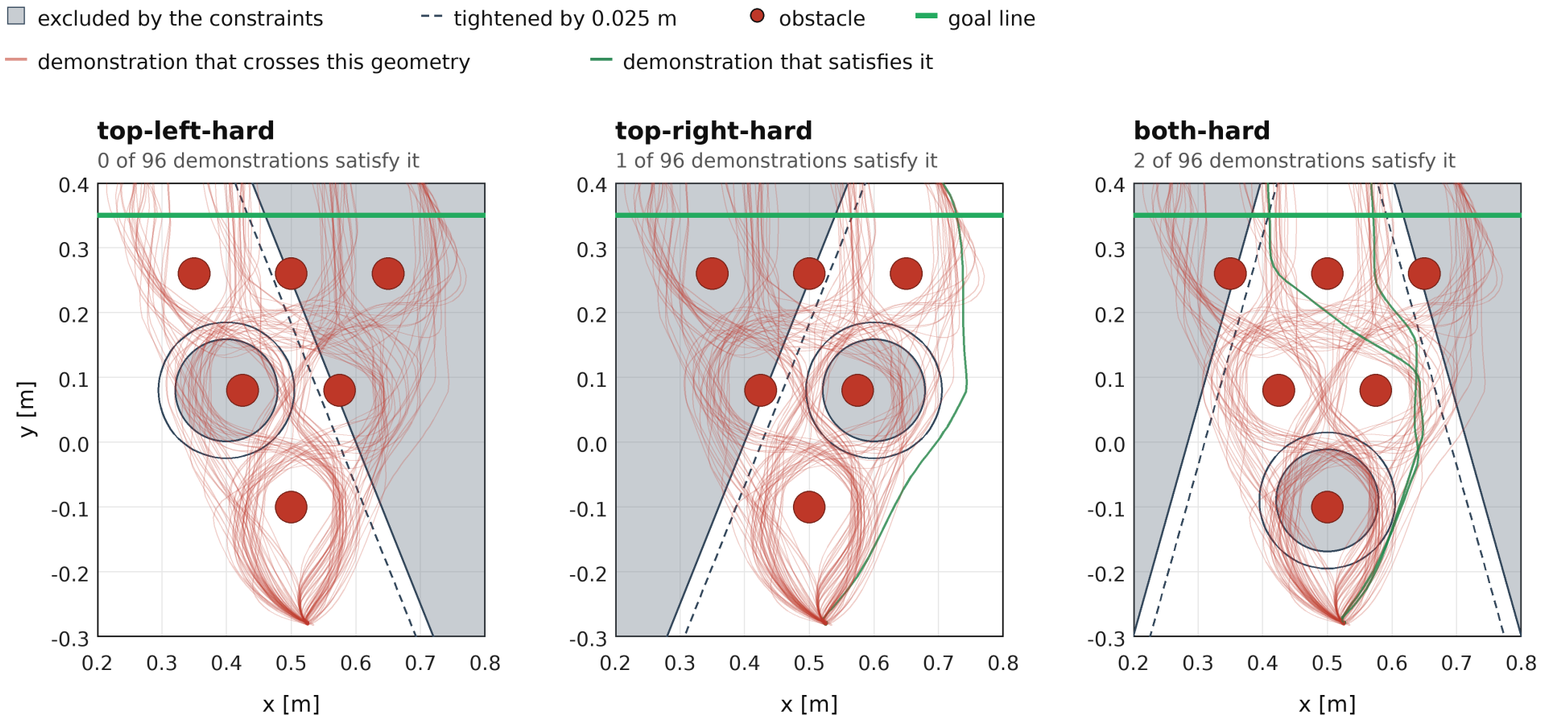


Illustration, not thesis data · animation A2

- **RQ1 — Generative model** (the sampling block)**.** Can the flow-based models reach the diffusion baseline's task success and constraint satisfaction with fewer sampling steps and less planning time per control step?
- **RQ2 — Constraints** (the projection inside it)**.** Does endpoint projection reach the goal with fewer constraint violations, fewer control steps or less computation than the per-step projection of DPCC?
- **RQ3 — Transfer** (what the loop observes and drives)**.** Do the answers to RQ1 and RQ2 hold when the observation becomes camera images, when the task requires aligning an object through contact, and when the platform is a quadrotor?

Three steps: **D3IL-avoiding**, the benchmark of DPCC unchanged → **D3IL-aligning**, observed through cameras → **quadrotor**, three scenes.

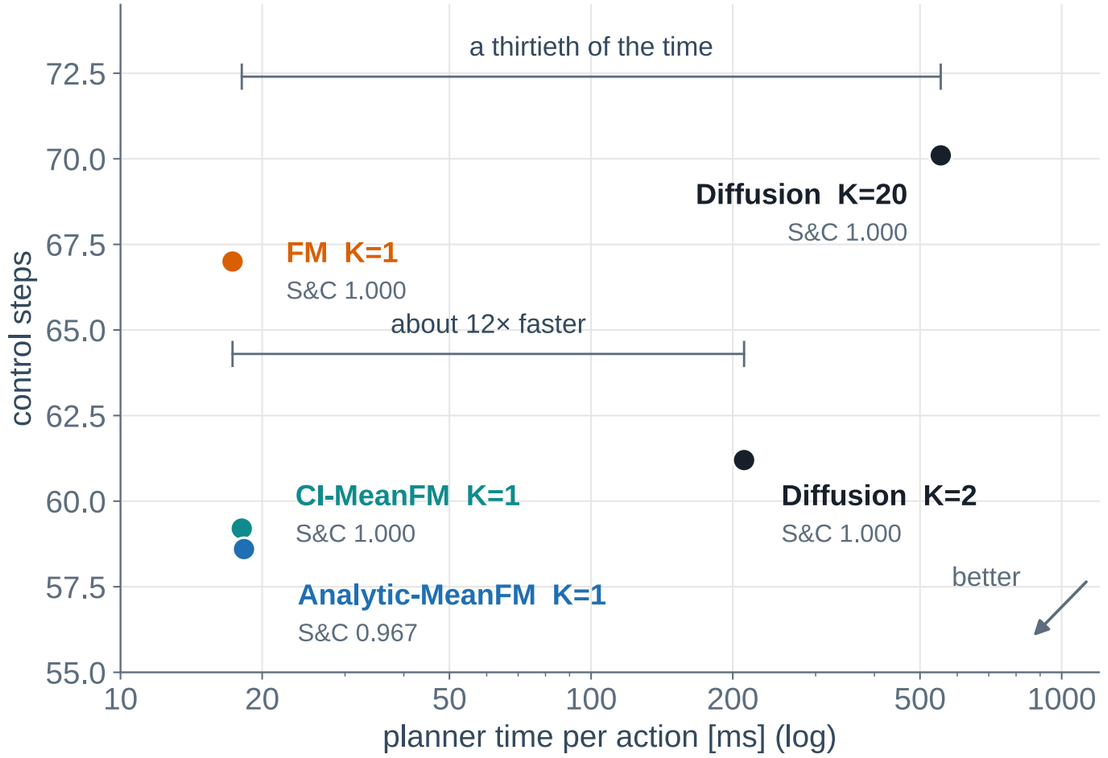
D3IL-avoiding: the benchmark of DPCC, unchanged



The 96 demonstrations against the three geometries. Thesis Fig. 5.9

- The scene of the animation, as DPCC evaluates it: three constraint geometries at test time, satisfied by 0, 1 and 2 of the 96 demonstrations.
- The same environment, data, projection and protocol as DPCC: five training seeds × two episodes per geometry.

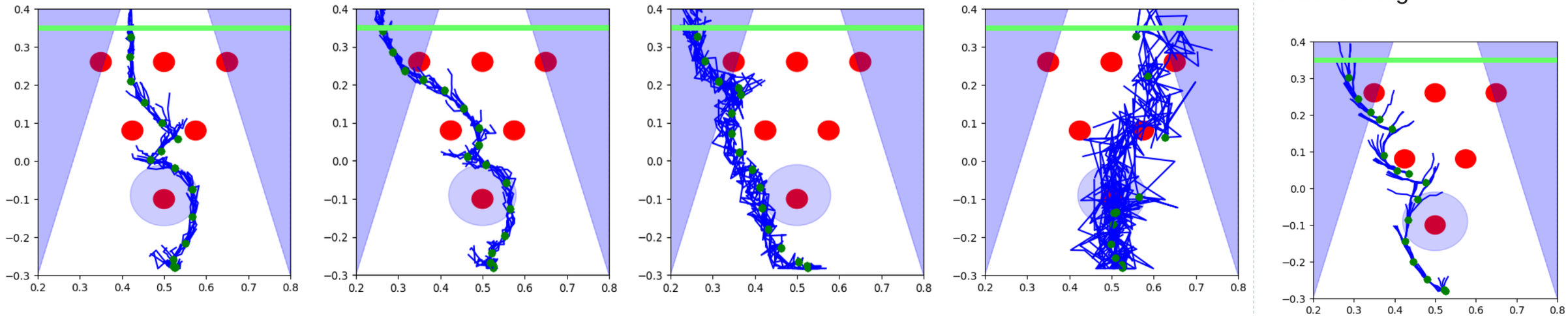
On the benchmark of DPCC, the average-velocity models reach the goal in the fewest control steps, at a thirtieth of its diffusion model's time



Thesis numbers, at the protocol of DPCC: planner time per action against control steps; S&C = success with constraint satisfaction; each model under its selected rule; the diffusion points are the reproduced baseline.

The time a projection takes depends on how far from feasible the plan it receives is

Analytic-MeanFM · $K = 1$ **CI-MeanFM** · $K = 1$ **FM** · $K = 1$ **Diffusion** · $K = 1$ **Diffusion** · $K = 20$, trained budget



6.0 violating steps

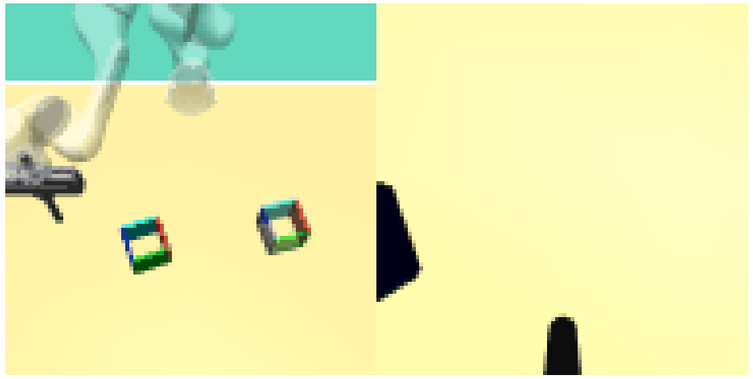
28.0 violating steps

18.5 violating steps

Plans without projection, every fourth control step, all four candidates; both-hard, one training seed; violating steps per episode in this cell. The executed path is not drawn.

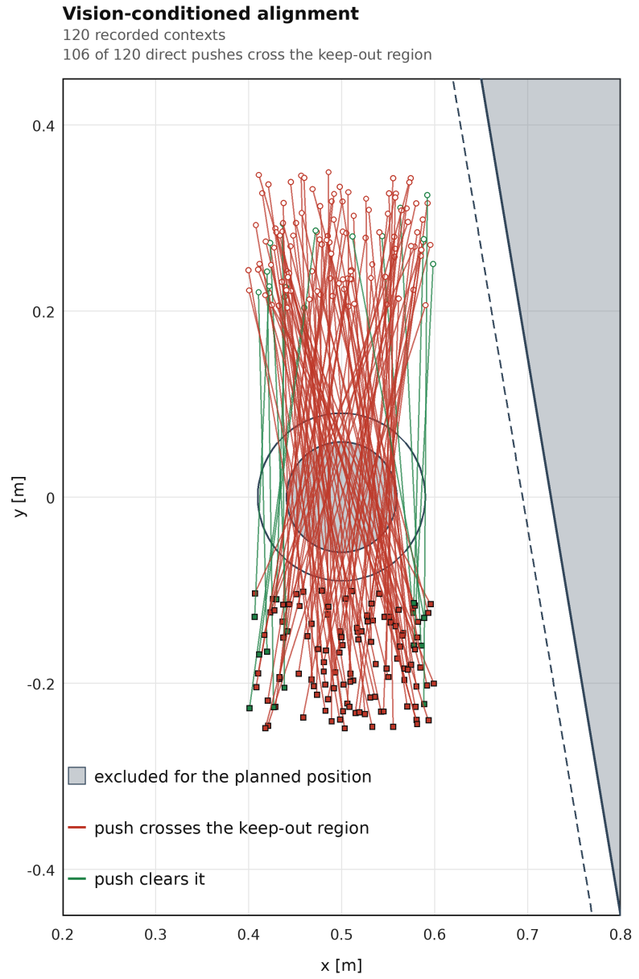
- At one evaluation Analytic-MeanFM's successive plans form a narrow ribbon; the diffusion baseline's plans jump across the workspace from one control step to the next.
- At its own twenty steps the baseline draws a single tight bundle, the upper half of which lies inside the left halfspace.

D3IL-aligning: the robot observes two camera views



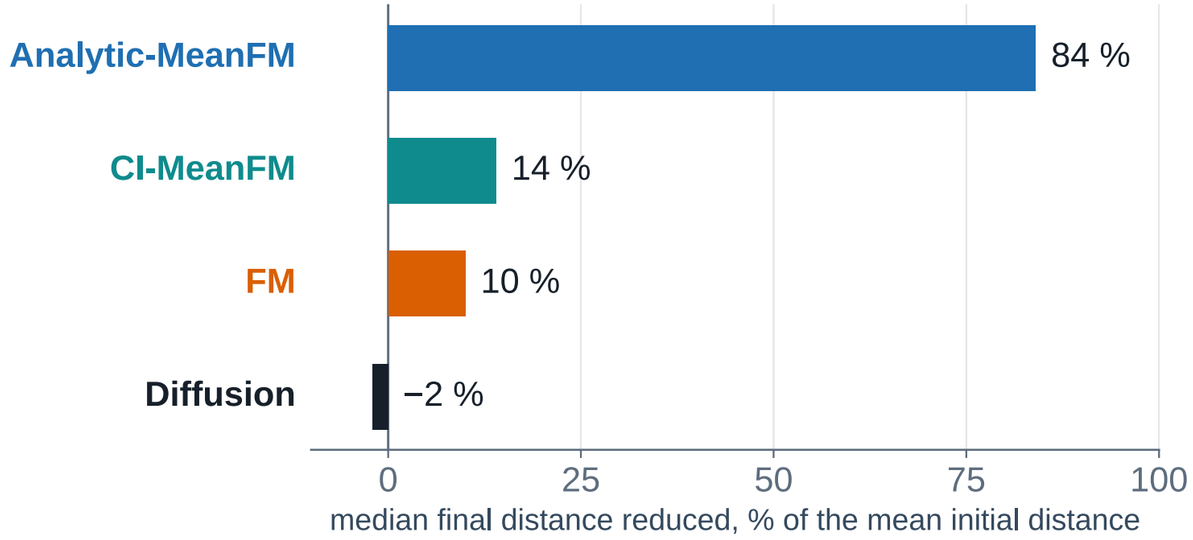
Expert demonstration (MuJoCo), the two camera views

- The manipulator pushes a box into a target pose; a context is an initial box pose and a target pose.
- No box or target pose in the observation vector: the policy must find the box in two RGB streams, a front and an in-hand view.
- The two views become one latent of 128 numbers, a ResNet-18 each, that enters every U-Net block.
- **The constraints**, right: a keep-out disk between box and target and a halfspace at the far corner. The straight push crosses the disk in 106 of 120 recorded contexts, 119 once the region is tightened.



The constraint set and the 120 recorded contexts. Thesis Fig. 5.10

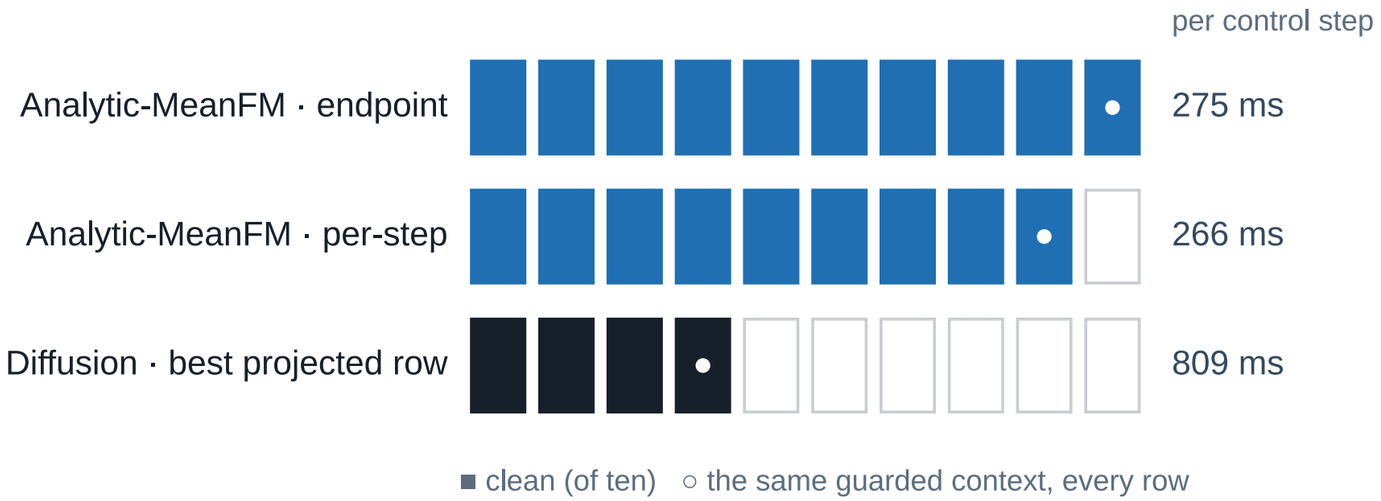
On D3IL-aligning the dominance repeats where the task separates the models



Thesis numbers: distance closed without projection at $K = 20$, ten training-set contexts, one seed; median final against mean initial distance.

- At two evaluations, at a seventh to a tenth of their cost, Analytic-MeanFM’s reduction is already two fifths.
- **CI-MeanFM** does not improve with the budget below a hundred evaluations.

At the budget this task needs, endpoint projection is the better projector on the constraint



Thesis numbers: contexts free of violations after projection, $K = 20$, random selection; every x of 10 includes the guarded context.

- Endpoint projection keeps more contexts violation-free than per-step projection in seven of nine comparisons.
- The constraint is paid for in distance: no context ends within the 1.8 cm of the task's own test, projected or not.

Endpoint: 20 integration steps + 3 lookaheads (NFE 23); for the average-velocity models both evaluations query the zero-interval head. Times are measured.

Below the plan: from the setpoint to the actuators



Quadrotor (UAV-corridor, UAV-s-curve)

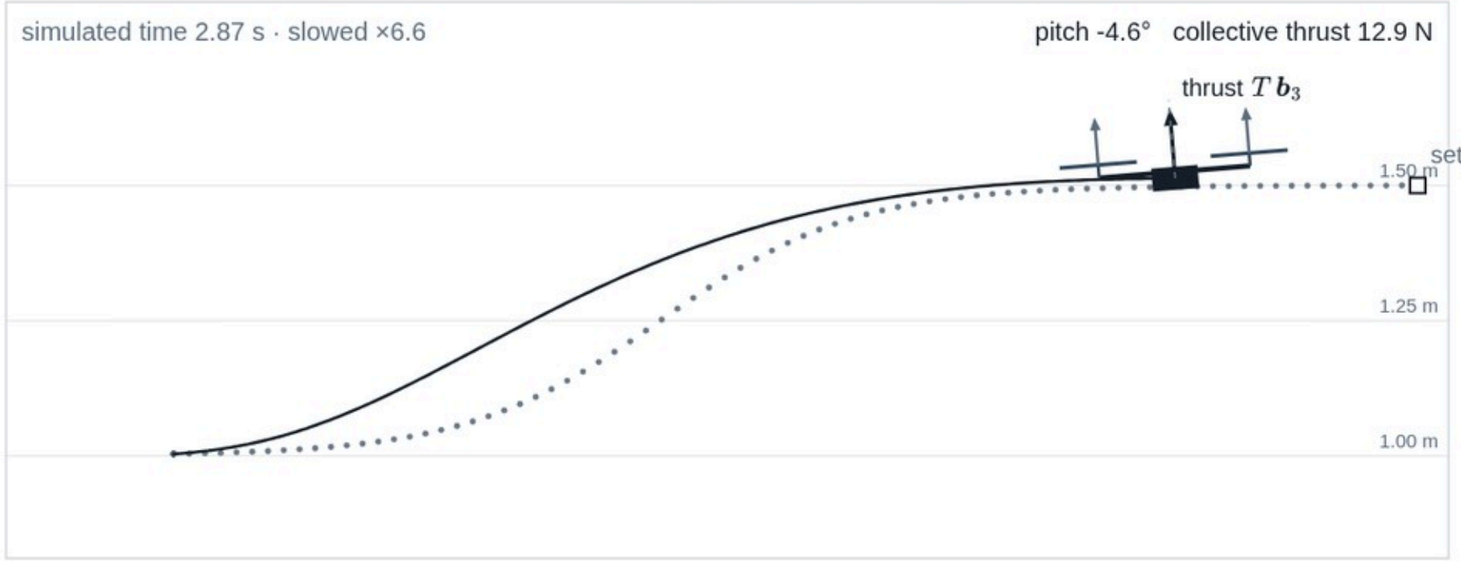
setpoint p_{des} → outer PD → force $\boldsymbol{F}$ → attitude $\boldsymbol{b}_{3,\mathrm{des}} = \boldsymbol{F}/\|\boldsymbol{F}\|$ → inner PD on $SO(3)$ → moment $\boldsymbol{\tau}_b$ → allocation → 4 rotor thrusts → MuJoCo integrates

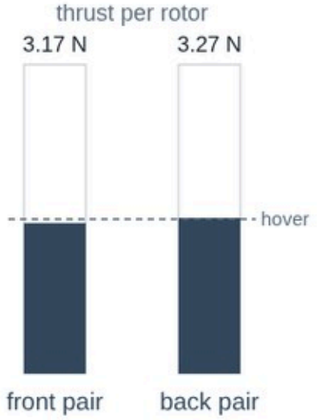
$$\boldsymbol{a}_{\mathrm{cmd}} = -\mathbf{K}_p\boldsymbol{e}_p - \mathbf{K}_d\boldsymbol{e}_v + \ddot{\boldsymbol{p}}_{\mathrm{des}}, \quad \boldsymbol{F} = m\,(\boldsymbol{a}_{\mathrm{cmd}} + g\,\boldsymbol{e}_3)$$

$$\boldsymbol{\tau}_b = -\mathbf{K}_R\boldsymbol{e}_R - \mathbf{K}_\omega\boldsymbol{e}_\omega + \boldsymbol{\omega}\times\mathbf{J}_b\boldsymbol{\omega}, \quad \mathbf{M}\boldsymbol{u} = \begin{bmatrix} T \\ \boldsymbol{\tau}_b \end{bmatrix}$$

Skydio X2 · Thesis Fig. 5.1 (MuJoCo render)

Gains of record: $\mathbf{K}_p = \mathrm{diag}(4,4,8)$, $\mathbf{K}_d = \mathrm{diag}(3,3,4)$, $\mathbf{K}_R = \mathrm{diag}(70,70,4)$, $\mathbf{K}_\omega = \mathrm{diag}(2.5,2.5,1)$; no integral term.



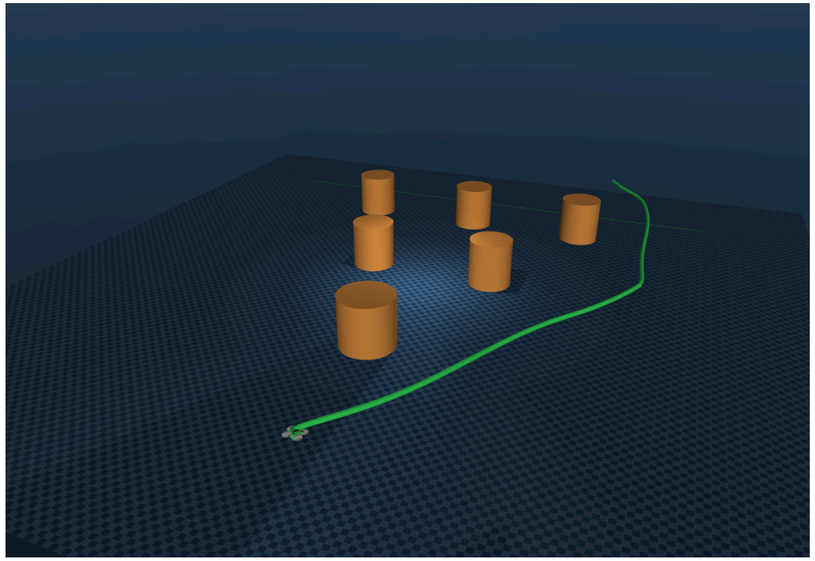




drawn: a planar rigid body with the X2's mass and pitch inertia, flown by this law (illustration)

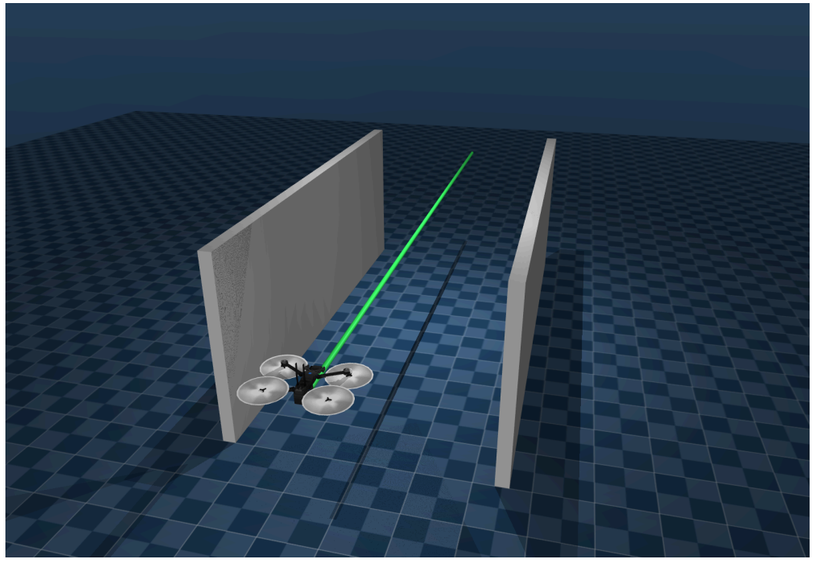
Illustration, not thesis data · D3IL-avoiding scene, toy planner on its 96 demonstrations

Quadrotor: three scenes that ask three different questions



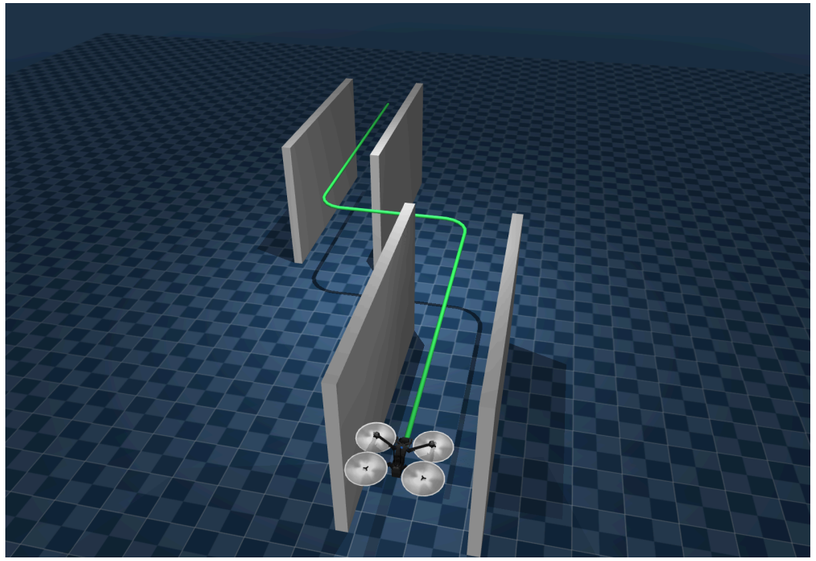
UAV-pillars

D3IL-avoiding flown by the quadrotor



UAV-corridor

constraints that leave the horizontal plane



UAV-s-curve

two sharp turns for the tracking controller

Skydio X2 of MuJoCo Menagerie, commanded through its rotor thrusts by a cascaded geometric tracking controller.

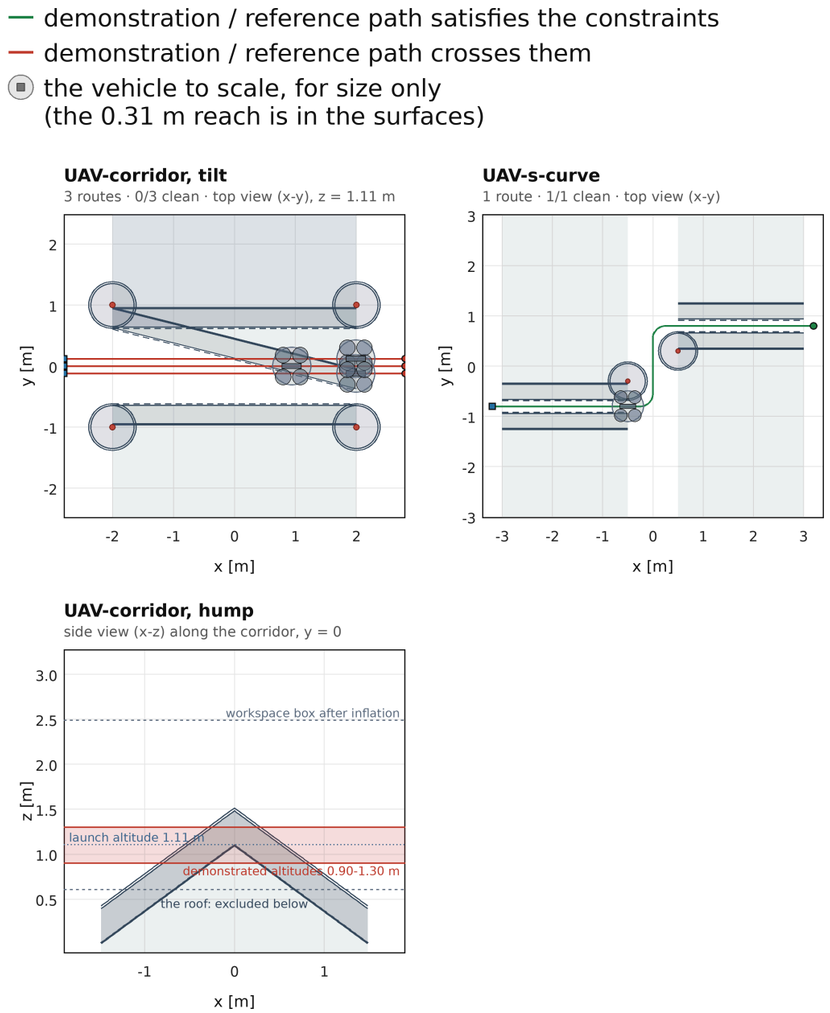
UAV-pillars: the result of D3IL-avoiding holds when its plans are flown



Thesis numbers: success within the constraints at the operating points, arm against quadrotor.

- After projection no flight violates the declared constraints; the vehicle tracks more closely than the arm.
- Every failure is a collision with one of the five pillars the constraint sets leave out.

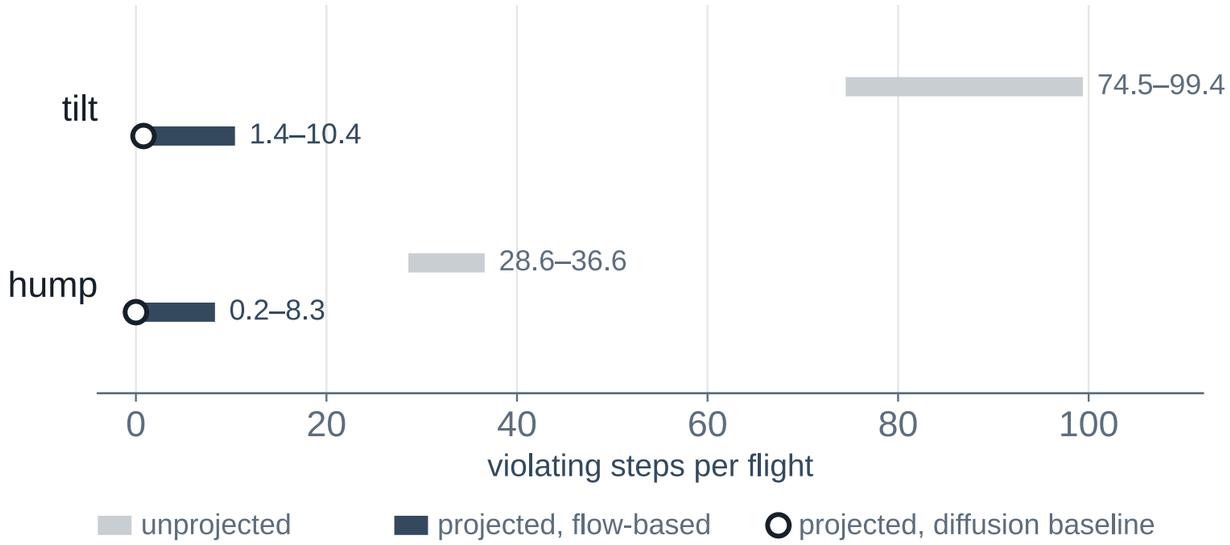
UAV-corridor: two test-time constraints leave the horizontal plane



- **Tilt:** one plane slopes by 14° across the corridor and leans 60° from the vertical, so it pushes the vehicle sideways and downward at once. At the launch altitude a sideways move alone does not clear it.
- **Hump:** a roof in the vertical plane, rising to 1.10 m at the middle of the corridor, which the vehicle must climb over.
- **Test-time only:** the models are trained on a narrower, level corridor. The three demonstrated lanes clear both walls, but every one crosses the tilt and passes under the hump.
- Each surface carries the rotor reach of 0.31 m; the two constraints are evaluated separately.

Left: UAV-corridor, the tilt from above and the hump from the side; right: UAV-s-curve. Thesis Fig. 5.11

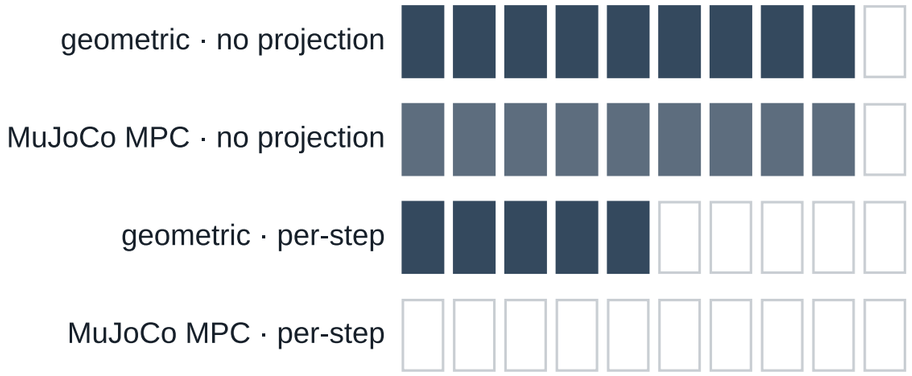
UAV-corridor: the projection scheme of DPCC carries over to three dimensions



Thesis numbers: violating steps per flight, before and after projection.

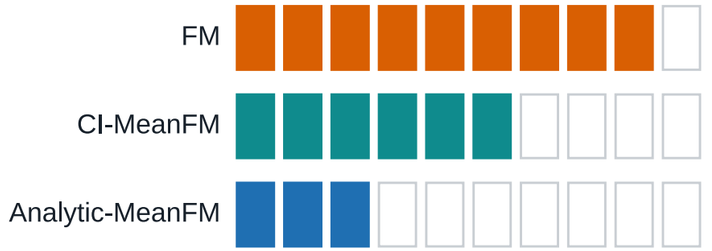
- Projected, the lowest altitude under the tilt drops on 540 of 560 flights and the highest over the hump rises on all 560; per-step projection at one and two evaluations stops short of the apex.
- The objectives cannot be told apart where the flights traverse the corridor at a jointly tested budget; the budget can; between the two projectors a trade-off.

UAV-s-curve: the controller decides whether a plan is flown



■ crossed the finish line (of ten) · 0 of 40 free of violations

Thesis numbers: flights that crossed the finish line, by controller and projection; none of the 40 is free of violations.



at one evaluation, no projection, cascaded geometric controller

By objective, at one evaluation, unprojected.

The controller’s cost, about 125 ms against 5 ms per control step, is read on the unprojected pair.

Limitations and outlook

Limitations

- A changed geometry asks for a new feasible set, not a new model; an obstacle that moves during an episode is not evaluated.
- The plan is a sequence of positions; below it is the plant’s controller.
- A constraint is what is declared: the projector protects nothing else.
- Every environment is a simulation.

Outlook

- Human demonstrations for the quadrotor scenes.
- A longer planning horizon than eight steps.
- Constraints from perception: depth or range sensing that emits the projector’s primitives.

Answers to the research questions

- **RQ1 — Yes, on the benchmark of DPCC.** With human demonstrations the average-velocity models come first; with the generated quadrotor demonstrations the objectives tie, or the simplest leads.
- **RQ2 — Where it has steps to guide.** Endpoint projection guides the sample only where the budget and the threshold leave a step before the terminal one; there it is the better projector on the constraint at a comparable planning time. It rarely shortens the task, and on the corridor it trades.
- **RQ3 — Where the demonstrations are of the same kind.** The answers hold on camera observations and contact, and the constraints transfer to the quadrotor; the plant and the demonstrations change the rest.

Across the environments

1. The time a projection takes depends on how far from feasible the plan it receives is.
2. A task saturated at one evaluation hides what a task that rewards more evaluations exposes.
3. The demonstrations bound what a model can show.

Thank you · Questions

End

References and protocol

Sources

- **DPCC**: Römer, von Rohr and Schoellig, *Diffusion Predictive Control with Constraints*, L4DC 2025, PMLR 283 (arXiv:2412.09342).
- **D3IL**: Jia et al., *Towards Diverse Behaviors*, ICLR 2024 (arXiv:2402.14606).
- **MeanFlow**: Geng, Deng, Bai, Kolter and He, *Mean Flows for One-step Generative Modeling*, 2025 (arXiv:2505.13447).
- **AlphaFlow**: Zhang et al., *AlphaFlow: Understanding and Improving MeanFlow Models*, 2025 (arXiv:2510.20771).
- **HardFlow**: Li, Alim and Azizan, *HardFlow: Hard-Constrained Sampling for Flow-Matching Models via Trajectory Optimization*, 2025 (arXiv:2511.08425v3).

Protocol, in short

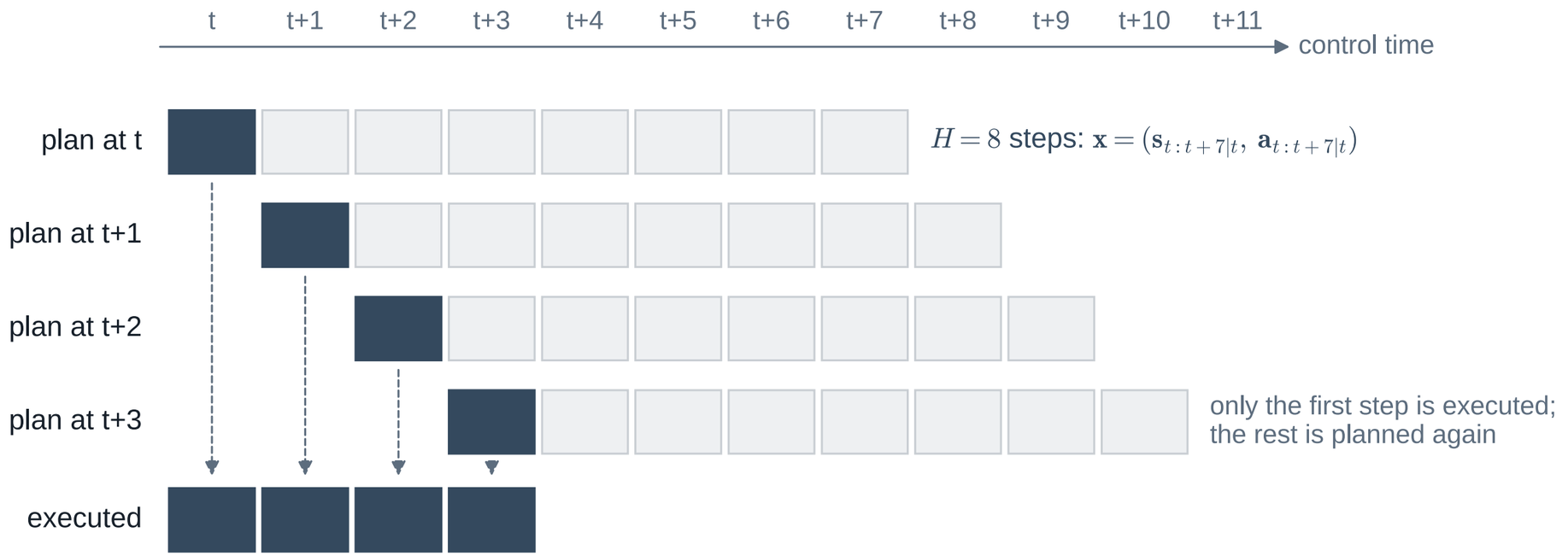
- **S&C**: success with constraint satisfaction; violations are counted on the executed trajectories against the nominal constraints.
- **Selection rules**: random (r), cumulative projection cost (c), temporal consistency (t); each model is reported under its selected rule.
- **Baseline**: the DPCC diffusion model reproduced on the thesis's node; its published result sits next to the reproduction (Table 5.6). Two details follow the released code: the displacement action and a 200-step limit.
- **Time**: the planner's time per action, projection included, on one GPU of the thesis's node; controller and simulator excluded.
- **Recipes**: backbone, data and evaluation are matched; each objective trains with its own recipe (Table 5.8).

Notation

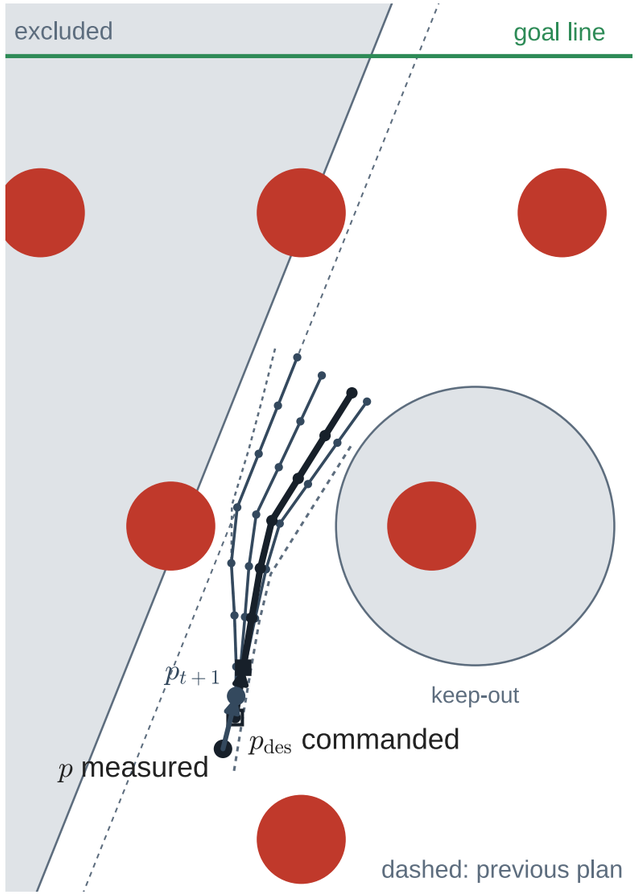
$\boldsymbol{x}$	a plan, 8 steps × d values, flattened
$\boldsymbol{x}^0 \dots \boldsymbol{x}^K$	diffusion's chain, clean plan to pure noise; k, K its steps
$\boldsymbol{x}_0,\ \boldsymbol{x}_1,\ \boldsymbol{x}_\tau$	flow's noise sample, data plan, and the point at time τ on the path between them
$\tau \in [0,1]$, $\tau_k = k/K$	generation time, 0 noise, 1 data; the Euler grid
$\bar{\boldsymbol{x}}_{k+1}$	the Euler step's result before projection
$\boldsymbol{\epsilon},\ \boldsymbol{z}$; $\beta_k,\ \alpha_k,\ \bar{\alpha}_k$; σ_k	Gaussian noise; diffusion's schedule and its products; the reverse step's noise level
$\boldsymbol{c}$, $\mathbf{e}$	the conditioning (observation); the 128-number image latent
θ	the network's parameters: a subscript marks a learned quantity
$\boldsymbol{\epsilon}_\theta$, $\boldsymbol{\mu}_\theta$	diffusion: the predicted noise; the reverse-step mean
v, v_θ, $\boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$	instantaneous velocity: exact, learned (FM), and the one training path's
$u(\boldsymbol{x}_r, r, h)$, u_θ	average velocity over $[r, r+h]$, the displacement over h; learned by Analytic-MeanFM and CI-MeanFM, h an input

u_{tgt}, sg	the average head's regression target; stop-gradient (a fixed teacher)
$\text{JVP}(u_\theta; \cdot, (\boldsymbol{v}, +1, -1))$	the exact derivative of u_θ along "move with $\boldsymbol{v}$, advance r, shrink h"
α, $\delta = \alpha h$	CI-MeanFM's consistency ratio and the straight first part; α from 1 to 0.2
$\mathcal{Z}_f$, $\Pi_{\mathcal{Z}_f}$	the feasible set; the projection onto it, and $\frac{1}{2}\|\Pi(\boldsymbol{x}) - \boldsymbol{x}\|^2$ its cost
K, η, n_{guide}, NFE	budget; activation threshold (last $\lceil \eta K \rceil$ steps); guiding steps $\lceil \eta K \rceil - 1$; evaluations $K + n_{\text{guide}}$
$\bar{\boldsymbol{x}}_1$, $\hat{\boldsymbol{x}}_1^\star$	the predicted final plan and its projection (endpoint projection)
$\mathbf{o}_t = (p_{\text{des},t}, p_t)$, Δp_t, B	observation; the plan's first increment; candidates per step (4)
S&C; r, c, t	success with constraint satisfaction; the selection rules random, cumulative cost, temporal consistency
$\mathbf{K}_p, \mathbf{K}_d, \mathbf{K}_R, \mathbf{K}_\omega$	the quadrotor controller's PD gains, outer and inner loop

Receding horizon: plan eight steps, execute the first, plan again



One control step, block by block, on D3IL-avoiding



Illustration, not thesis data

1 · Observation — the commanded and the measured position, $\mathbf{o}_t = (p_{\text{des},t},\, p_t)$.

2 · Candidates — $B = 4$ plans, each drawn from independent noise.

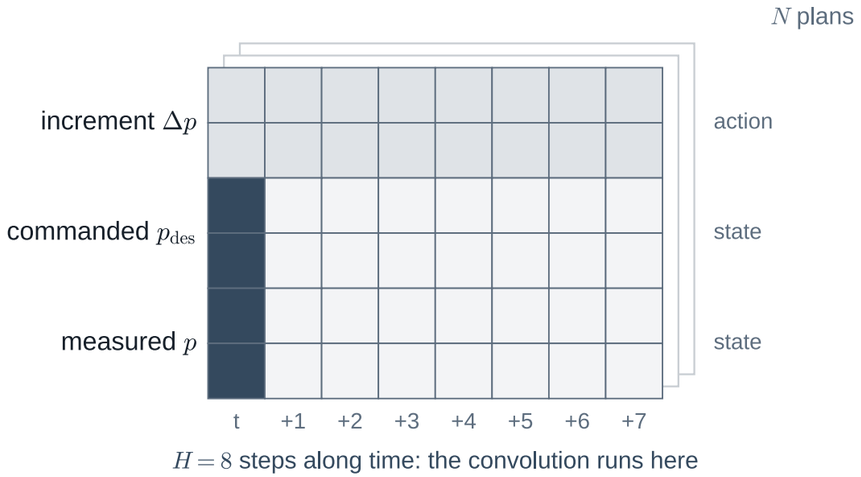
3 · Sampling with projection — the model's K steps; the projection constrains the iterates or the predicted endpoint.

4 · Candidate selection — temporal consistency: the candidate closest to the previous plan.

5 · First increment → setpoint — $p_{\text{des},t+1} = p_{\text{des},t} + \Delta p_t$.

6 · Tracking controller — differential IK → joint PD moves the arm towards the setpoint; the next observation starts the next step.

The network reads a plan as an 8 × d signal



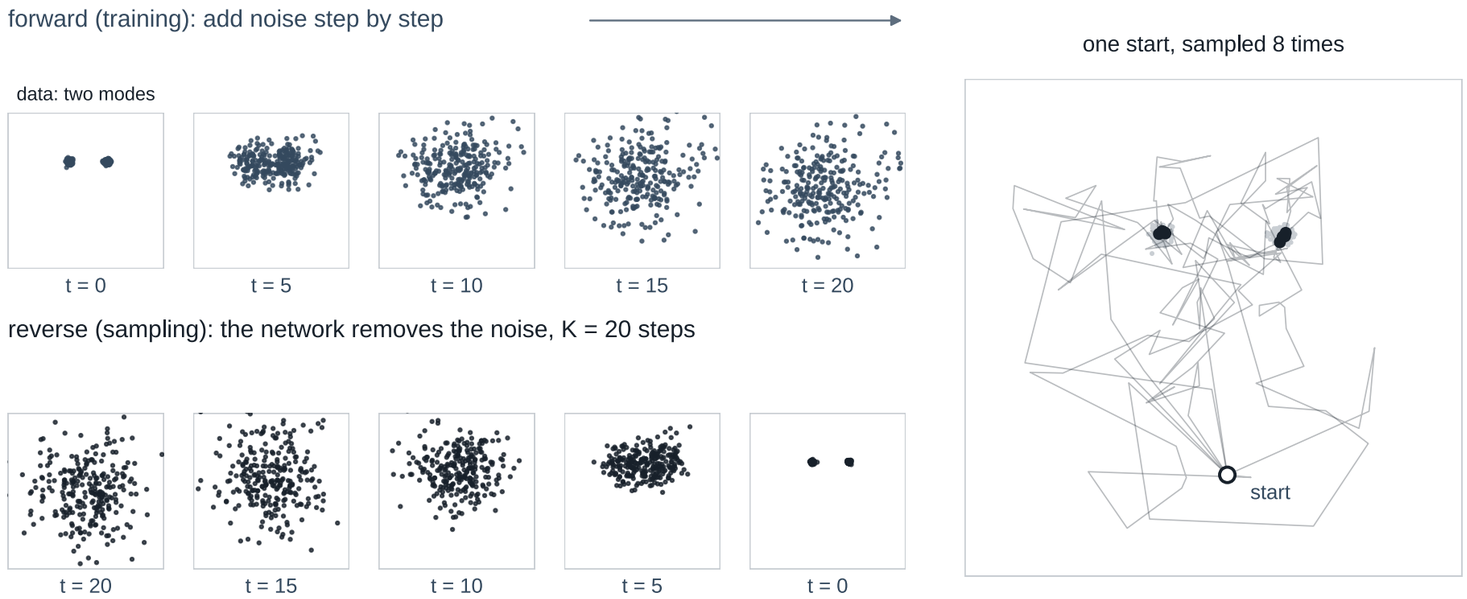
The plan tensor of D3IL-avoiding, $(N, 8, 6)$. Dark: the known current state, imposed by inpainting and masked in the loss.

Environment	Plan in / out	State anchor
D3IL-avoiding	$(N, 8, 6)$	$(N, 4)$
D3IL-aligning	$(N, 8, 9)$	$(N, 6)$
UAV-corridor, UAV-s-curve	$(N, 8, 12)$	$(N, 9)$

D3IL-aligning adds two RGB views, $2 \times (N, 1, 3, 96, 96)$, encoded to $(N, 128)$. UAV-pillars flies the D3IL-avoiding models.

- A temporal U-Net: convolution along time, channel widths $(d, w, 2w, 4w, 8w)$, three stride-two stages ($8 \to 4 \to 2 \to 1$) with skip connections.
- Conditioning is a per-channel bias in every block: the time embedding of τ, plus h for Analytic-MeanFM and CI-MeanFM, plus the image latent on D3IL-aligning.
- The output has the plan's shape: v_θ, u_θ, or the baseline's noise $\boldsymbol{\epsilon}_\theta$.

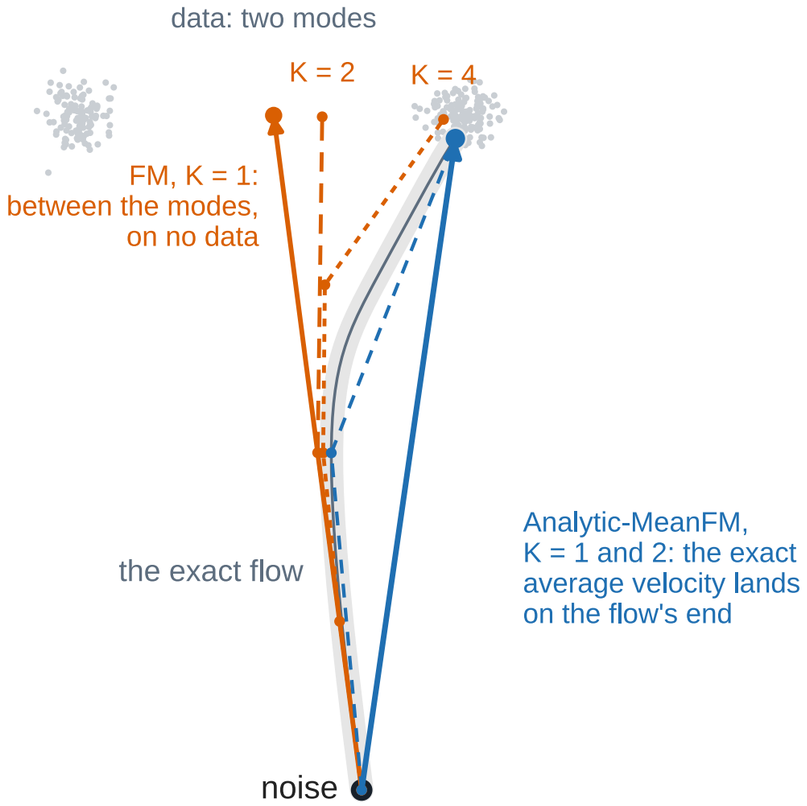
Diffusion: predict the noise, remove it step by step



- A network is trained to predict the noise added at each step; generation starts from pure noise and removes it step by step.
- The number of steps is tied to the training schedule: fixed when training ends ($K = 20$ in DPCC); fewer steps need a different sampler.
- The sampler adds noise at every step but the last: plans for the same observation can differ.

Illustration, not thesis data

Flow matching: a velocity field transports noise to data



	learns	steps K	sampling
Diffusion (DPCC's sampler)	the noise	fixed at training	adds noise
FM	velocity v	chosen at deployment	deterministic given the noise
Analytic-MeanFM, **CI-MeanFM**	average velocity u	chosen at deployment	deterministic given the noise

- Straight per-sample paths average into a curved field, and coarse steps along a curved field are inaccurate.
- One evaluation of an accurate average velocity covers the whole interval.

Illustration, not thesis data

Diffusion: train by adding noise step by step, sample by removing it step by step

Train · forward, one step at a time: $\boldsymbol{x}^0 \to \boldsymbol{x}^1 \to \cdots \to \boldsymbol{x}^K$. Each step adds a little Gaussian noise with the schedule's β_k; after K steps nothing of the data is left.

$$\boldsymbol{x}^k = \sqrt{1-\beta_k}\,\boldsymbol{x}^{k-1} + \sqrt{\beta_k}\,\boldsymbol{\epsilon}_k, \qquad \boldsymbol{\epsilon}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad k = 1, \dots, K$$

so that any step comes from the data in one shot:

$$\boldsymbol{x}^k = \sqrt{\bar{\alpha}_k}\,\boldsymbol{x}^0 + \sqrt{1-\bar{\alpha}_k}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Train · the network: given a noised plan and its step k, predict the noise $\boldsymbol{\epsilon}$ that took $\boldsymbol{x}^0$ to $\boldsymbol{x}^k$.

$$\mathcal{L}(\theta) = \mathbb{E}_{k,\,\boldsymbol{x}^0,\,\boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}^k, k) \right\|^2$$

Sample · reverse, one step at a time: $\boldsymbol{x}^K \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \to \boldsymbol{x}^{K-1} \to \cdots \to \boldsymbol{x}^0$. Each step removes the predicted noise and adds a little fresh noise (σ_K … σ_1; none at the last step).

$$\boldsymbol{x}^{k-1} = \frac{1}{\sqrt{\alpha_k}} \left(\boldsymbol{x}^k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\,\boldsymbol{\epsilon}_\theta(\boldsymbol{x}^k, k) \right) + \sigma_k \boldsymbol{z}, \quad \boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad k = K, \dots, 1$$

$\alpha_k = 1 - \beta_k$, $\bar{\alpha}_k = \prod_{i \le k} \alpha_i$. The schedule is set before training, so K is fixed by it: $K = 20$ in DPCC. DPCC projects the iterate between the reverse steps.

Flow matching: train a velocity field, sample by integrating it from noise to data

Train · the velocity field: pair a noise sample $\boldsymbol{x}_0$ with a data sample $\boldsymbol{x}_1$; the straight path between them has velocity $\boldsymbol{x}_1 - \boldsymbol{x}_0$ at every time. Draw τ, and regress the network onto that velocity.

$$\boldsymbol{x}_\tau = (1-\tau)\,\boldsymbol{x}_0 + \tau\,\boldsymbol{x}_1, \qquad \frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = \boldsymbol{x}_1 - \boldsymbol{x}_0, \qquad \mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{\tau,\boldsymbol{x}_0,\boldsymbol{x}_1}\left\| v_\theta(\boldsymbol{x}_\tau,\tau) - (\boldsymbol{x}_1 - \boldsymbol{x}_0)\right\|^2$$

Sample · integrate the field: the learned v_θ defines an ODE; a plan is its solution from a noise sample at $\tau = 0$ to $\tau = 1$.

$$\frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = v_\theta(\boldsymbol{x}_\tau,\tau), \quad \boldsymbol{x}_0 \sim \mathcal{N}(\mathbf{0},\mathbf{I}) \qquad \Longrightarrow \qquad \boldsymbol{x}_1 = \boldsymbol{x}_0 + \int_0^1 v_\theta(\boldsymbol{x}_\tau,\tau)\,\mathrm{d}\tau$$

In practice, Euler: the integral is replaced by K steps of size $1/K$ along the field; K is chosen at deployment, not at training.

$$\boldsymbol{x}_{\tau_{k+1}} = \boldsymbol{x}_{\tau_k} + \tfrac{1}{K}\, v_\theta(\boldsymbol{x}_{\tau_k},\tau_k), \qquad \tau_k = \tfrac{k}{K}, \quad k = 0,\dots,K-1$$

Average velocity (what the MeanFM models learn instead): the mean of v over an interval of length h, which is the displacement over it divided by h; one accurate evaluation covers the whole interval.

$$u(\boldsymbol{x}_r, r, h) = \frac{1}{h}\int_r^{r+h} v(\boldsymbol{x}_\varsigma,\varsigma)\,\mathrm{d}\varsigma = \frac{\boldsymbol{x}_{r+h} - \boldsymbol{x}_r}{h} \qquad \boldsymbol{x}_{\tau_{k+1}} = \boldsymbol{x}_{\tau_k} + \tfrac{1}{K}\, u_\theta\big(\boldsymbol{x}_{\tau_k},\tau_k,h{=}\tfrac{1}{K}\big)$$

Three objectives inside the same constrained controller

FM

instantaneous-velocity matching

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}\left\| v_\theta\big((1-\tau)\boldsymbol{x}_0 + \tau\boldsymbol{x}_1,\ \tau,\ \boldsymbol{c}\big) - (\boldsymbol{x}_1 - \boldsymbol{x}_0)\right\|_2^2$$

Analytic-MeanFM

analytic average-velocity matching

$$u_{\mathrm{tgt}} = \mathrm{sg}\Big[\boldsymbol{v} + h\,\mathrm{JVP}\big(u_\theta;\ (\boldsymbol{x}_r, r, h),\ (\boldsymbol{v}, +1, -1)\big)\Big]$$

CI-MeanFM

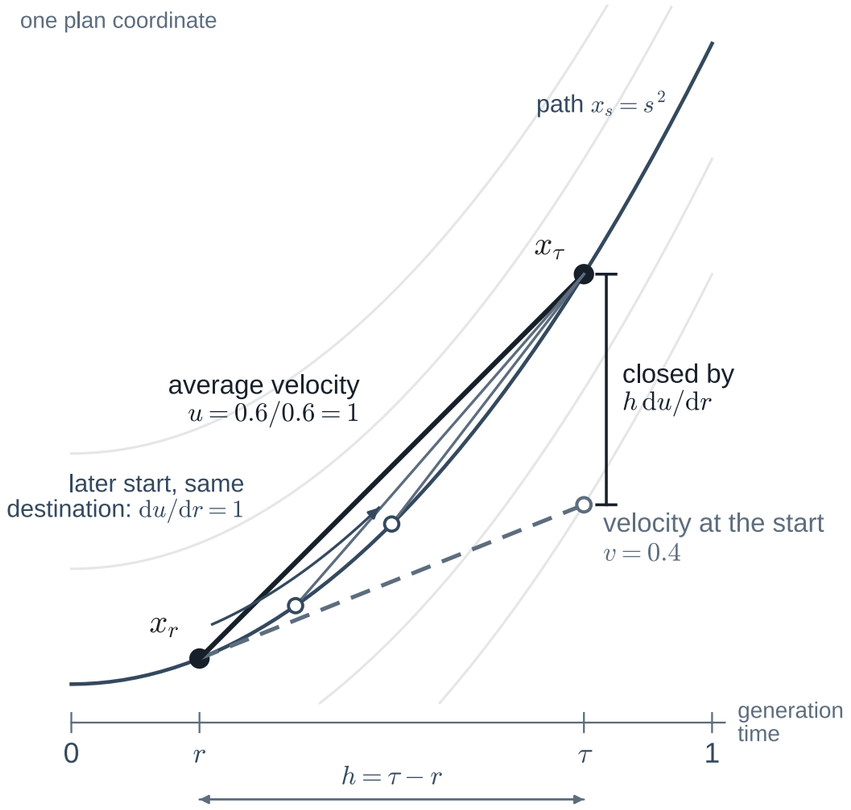
consistency-interpolated average-velocity matching

$$u_{\mathrm{tgt}} = \alpha\boldsymbol{v} + (1-\alpha)\,\mathrm{sg}\big(u_\theta(\boldsymbol{x}_r + \delta\boldsymbol{v},\ r+\delta,\ h-\delta)\big), \quad \delta = \alpha h$$

$\boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$; sg: stop-gradient. Backbone, projection, data and evaluation protocol are kept; the number of network evaluations becomes a choice made at deployment. Both average-velocity objectives train half of their samples at $h = 0$, where both targets are $\boldsymbol{v}$, and an auxiliary instantaneous head. Each objective trains with its own converged recipe (thesis Table 5.8): the results are those of the trained configurations.

	Analytic-MeanFM	CI-MeanFM
the target	the identity's derivative, taken exactly (a JVP)	whole interval = step + rest; α interpolates
loss weight, $h > 0$	1	α
curriculum	none	α from 1 by a sigmoid, held at 0.2

Analytic: the target comes from differentiating the average-velocity identity



One path, $x_s = s^2$: generation time across, one plan coordinate up, from $r = 0.2$ to $\tau = 0.8$.

Average velocity: the displacement over the interval, divided by its length. Here 1, while the velocity at the start is 0.4.

$$h\,u(\boldsymbol{x}_r, r, h) = \boldsymbol{x}_{r+h} - \boldsymbol{x}_r$$

Keep the destination, move the start: the destination does not change, so its derivative is zero.

$$0 = \boldsymbol{v} - u + h\,\frac{\mathrm{d}u}{\mathrm{d}r} \quad \Longrightarrow \quad u = \boldsymbol{v} + h\,\frac{\mathrm{d}u}{\mathrm{d}r}$$

Check: $1 = 0.4 + 0.6 \cdot 1$; the derivative term closes the gap. It moves all three inputs at once:

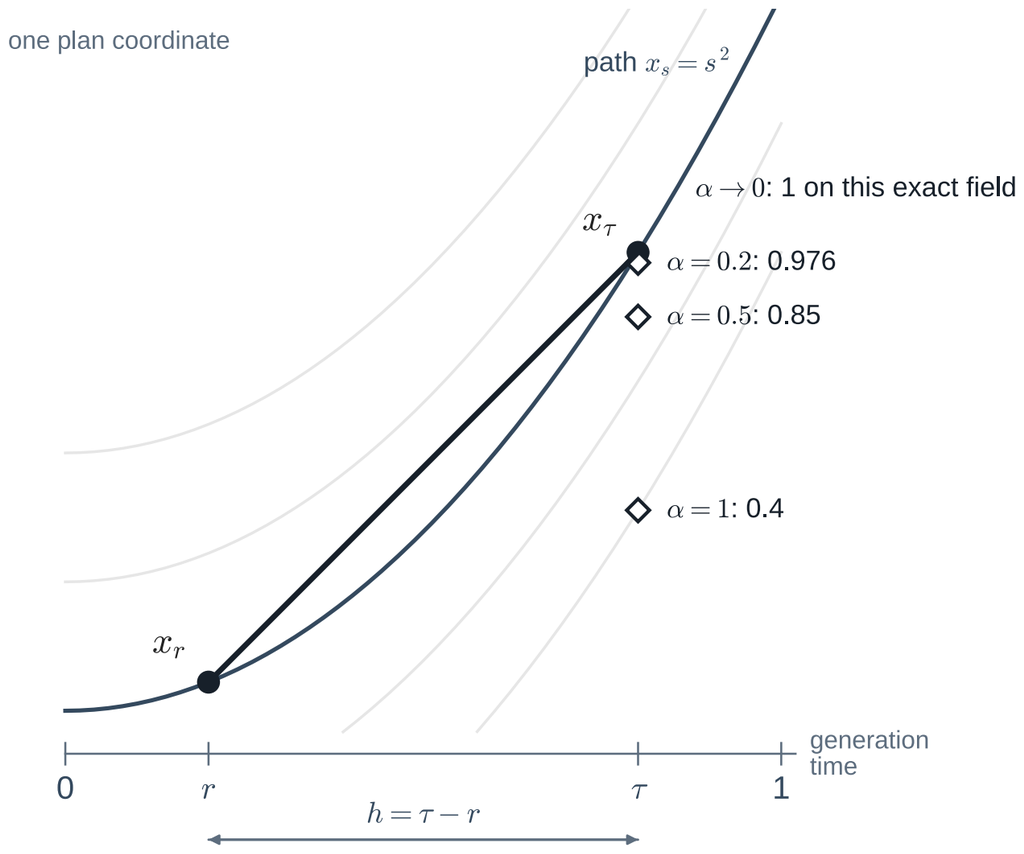
$$\frac{\mathrm{d}u}{\mathrm{d}r} = (\partial_{\boldsymbol{x}} u)\,\boldsymbol{v} + \partial_r u - \partial_h u$$

The analytic target puts the network in place of the unknown field, takes its exact derivative and freezes the result:

$$u_{\mathrm{tgt}} = \mathrm{sg}\left[\boldsymbol{v} + h\,\mathrm{JVP}\big(u_\theta;\ (\boldsymbol{x}_r, r, h),\ (\boldsymbol{v}, +1, -1)\big)\right]$$

Illustration, not thesis data

Consistency-interpolated: a finite step replaces the derivative, and α sets its length



Same path. The target is built in two parts: a straight step $\delta = \alpha h$ with $\boldsymbol{v}$, then the network's own prediction for the rest, frozen:

$$h\,u_{\text{tgt}} = \delta \boldsymbol{v} + (h - \delta)\ \text{sg}\left(u_\theta(\boldsymbol{x}_r + \delta \boldsymbol{v},\ r + \delta,\ h - \delta)\right)$$

α = 1: one straight step over the whole interval. The target is $v = 0.4$, the instantaneous velocity.

α = 0.5: a step of 0.3, then the prediction from the shifted point: 0.85.

α = 0.2: 0.976. In training, α follows a sigmoid schedule from 1 and is held at 0.2.

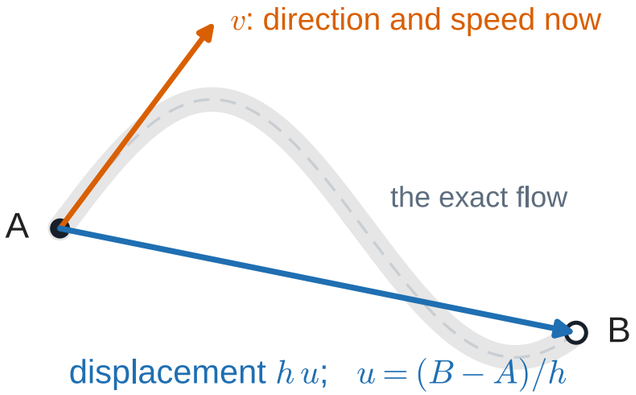
α → 0: the target tends to the network's current output, and its residual divided by α to the analytic residual: on this exact field, 1 and 0. At exactly α = 0 the implementation switches to the JVP target.

Consistency: the prediction for the whole interval must agree with a step plus the network's own prediction for the rest. **Interpolated:** α moves the target between the instantaneous velocity and that prediction.

Illustration, not thesis data: how the target is built, not how well it trains

A car on a winding road: what the average-velocity objectives learn

1 · velocity and average velocity

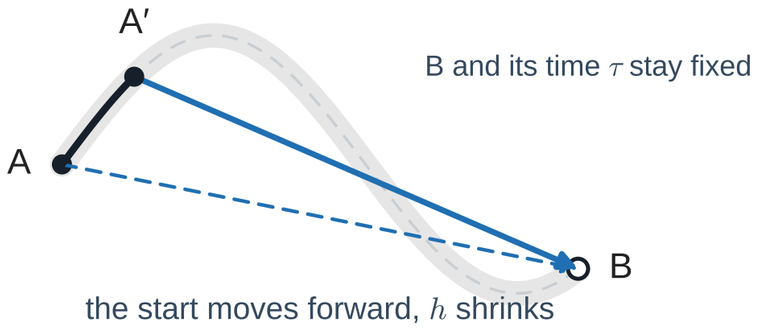


$$h\,u(\boldsymbol{x}_r, r, h) = \boldsymbol{x}_{r+h} - \boldsymbol{x}_r$$

one step from the start:

$\hat{\boldsymbol{x}}_1 = \boldsymbol{x}_0 + u_\theta(\boldsymbol{x}_0, 0, 1)$

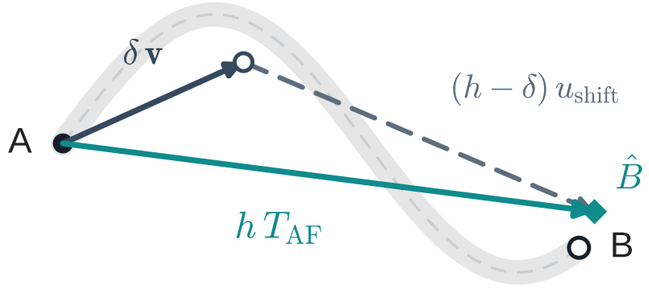
2 · Analytic-MeanFM: keep the destination



$$u = v + h\,\frac{\mathrm{d}}{\mathrm{d}r}u$$

target $\mathrm{sg}(\boldsymbol{v} + h\,\mathrm{JVP})$: a derivative at the current input

3 · CI-MeanFM: a two-part target



$$h\,u_{\mathrm{tgt}} = \delta\boldsymbol{v} + (h - \delta)\,u_{\mathrm{shift}}$$

$\delta = \alpha h$; a prediction at a shifted input; α from 1, held at 0.2 in training

Illustration, not thesis data: the road is the flow, h is generation time

Training: one average-velocity network, half of its samples anchored at h = 0

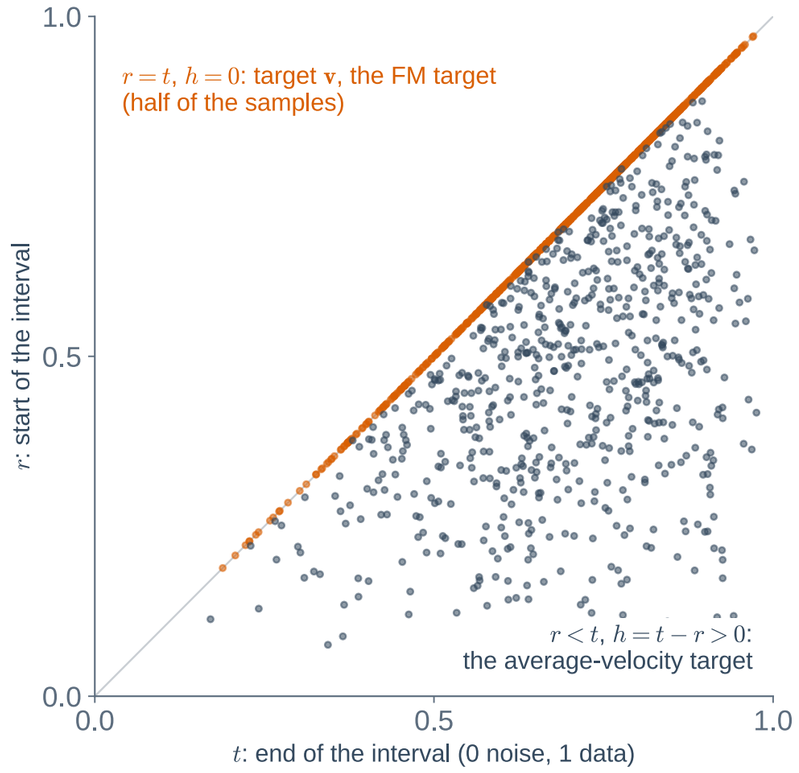
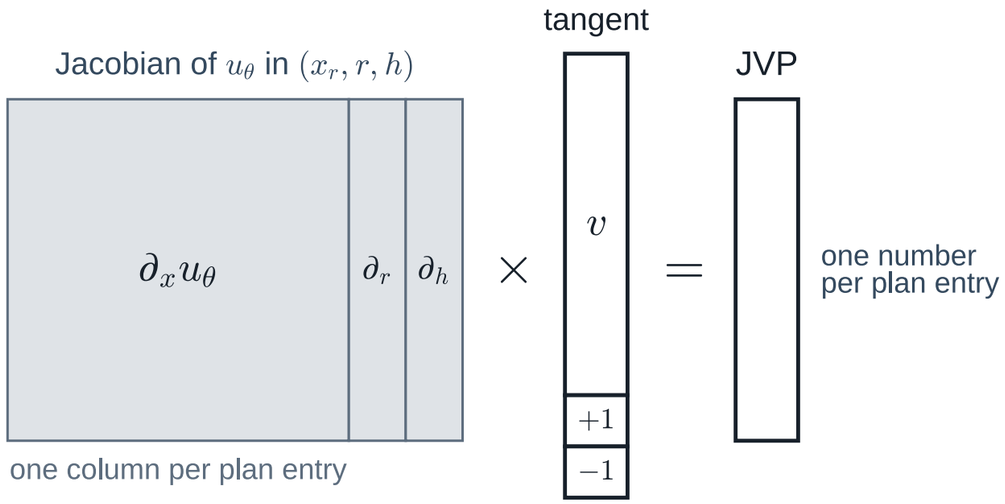


Illustration: 1400 draws from the thesis's time law, not training logs

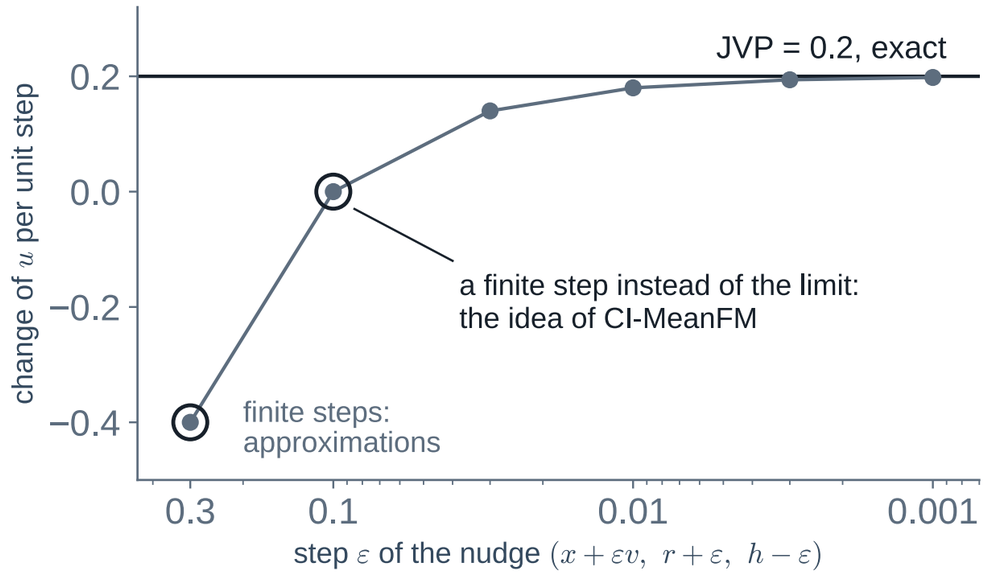
sample	target of the average head u_θ
$h = 0$, half of the samples	$\boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, the FM target
$h > 0$, **Analytic-MeanFM**	$\mathrm{sg}\big[\boldsymbol{v} + h\,\mathrm{JVP}\big]$
$h > 0$, **CI-MeanFM**	$\alpha\boldsymbol{v} + (1-\alpha)\,\mathrm{sg}\big(u_\theta(\cdot)\big)$, weighted by α

- A second, auxiliary head v_θ regresses $\boldsymbol{v}$ on every sample; sampling uses u_θ only.
- The two objectives share the network, its auxiliary head, the zero-interval samples and the sampler; they differ in how the average-velocity target is built, in its loss weight and in the curriculum (α from 1, held at 0.2).

The derivative is one Jacobian–vector product, computed exactly



Forward-mode differentiation gives the product in one pass; the Jacobian itself is never formed.



- **Worked example:** $u(x, r, h) = xh + r^2$ at $(2,\ 0.5,\ 0.4)$ with $v = 3$. The partials are 0.4, 1 and 2, so $\mathrm{JVP} = 0.4 \cdot 3 + 1 - 2 = 0.2$ and the target is $3 + 0.4 \cdot 0.2 = 3.08$.

- **Not** a derivative in the weights, which stop-gradient keeps out of the update; **not** used when sampling; its tangent $\boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$ is **not** the auxiliary head. The camera encoder stays outside the differentiated function.

Worked example from the thesis review

Analytic and consistency-interpolated: the same average velocity, two ways to build its target

Analytic-MeanFM: the network's derivative at the input

input	→	network	→	**exact derivative along**	→	target	→	regress
$(\boldsymbol{x}_r, r, h)$		u_θ		$(\boldsymbol{v}, +1, -1)$		$\boldsymbol{v} + h\,\mathrm{JVP}$		u_θ to it, frozen

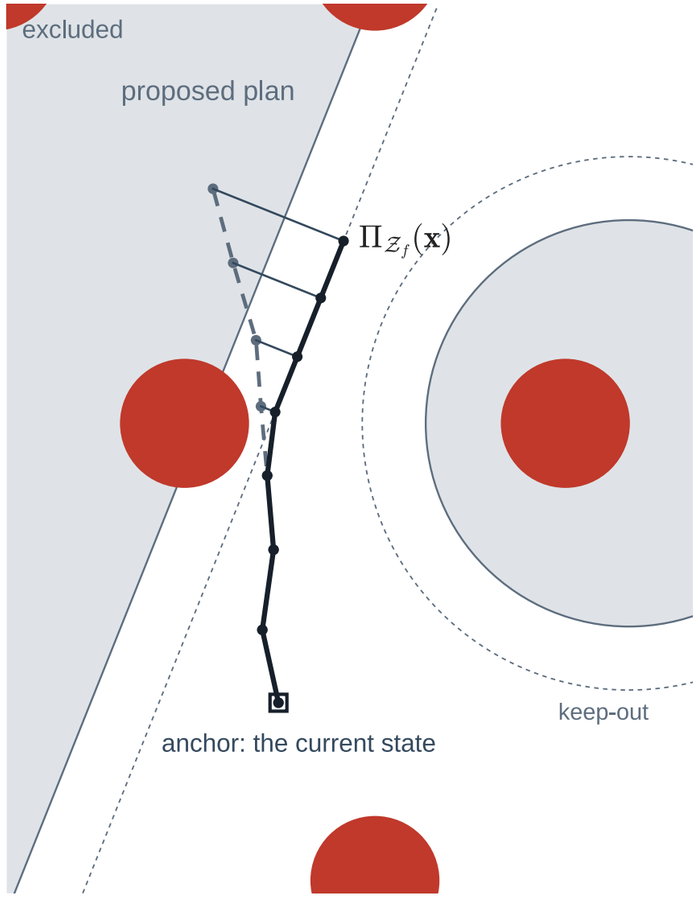
CI-MeanFM: the network's output at a shifted input

input	→	straight step	→	**network at the shifted input**	→	target	→	regress
$(\boldsymbol{x}_r, r, h)$		$\delta = \alpha h$ with $\boldsymbol{v}$				$\alpha\boldsymbol{v}$ $+(1-\alpha)\,u_{\mathrm{shift}}$		u_θ to it, frozen, weight α

	Analytic-MeanFM	CI-MeanFM
the name	the identity's derivative, taken exactly	whole interval = step + rest; α interpolates
loss weight, $h > 0$	1	α
curriculum	none	α from 1 by a sigmoid, held at 0.2

Shared: the network with its auxiliary instantaneous head and loss, the zero-interval samples, the predicted quantity and the sampler.

The projection: the closest plan that satisfies the constraints



Illustration, not thesis data; the dynamics rows are not drawn

$$\Pi_{\mathcal{Z}_f}(\mathbf{x}) = \arg\min_{\mathbf{z}} \ \frac{1}{2} \left\|\mathbf{z} - \mathbf{x}\right\|^2$$

- subject to the **dynamics rows** of the first-order model, the **bounds and halfspaces**, and one **quadratic keep-out inequality** per horizon step and obstacle. The first block is fixed at the measured state.
- The constraints are **tightened** by a margin for the model mismatch.
- **Goal-independent:** its objective is proximity to the proposed plan. Success is not its objective.
- A non-linear program, since the keep-out rows are non-convex, solved by SLSQP. Its cost $\frac{1}{2}\left\|\Pi_{\mathcal{Z}_f}(\mathbf{x}) - \mathbf{x}\right\|^2$ is what the cumulative-cost selection rule sums.
- **A local numerical solve:** at most 1000 iterations; it can return its last iterate, so feasibility of the output is not guaranteed. Violations are counted on the executed trajectories.

Two ways to put the projection into sampling

Per-step projection (DPCC): project the current iterate

iterate $\mathbf{x}_{\tau_k}$	→	model step $\bar{\mathbf{x}}_{k+1}$	→	*no prediction of the final plan*	→	**project the iterate** $\Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_{k+1})$	→	*no pull-back: the projection is kept*	→	next iterate $\mathbf{x}_{\tau_{k+1}}$

Guidance. A feasibility likelihood around $\mathcal{Z}_f$, expanded to first order, gives a step to the projection; in the diffusion sampler $\mathbf{x}^{k-1} = \Pi_{\mathcal{Z}_f}(\boldsymbol{\mu}_\theta + \tilde{\sigma}_k \boldsymbol{\epsilon}^k)$. Early in sampling the iterate is still far from the data, so the keep-out rows are evaluated on positions the sampler has not yet produced.

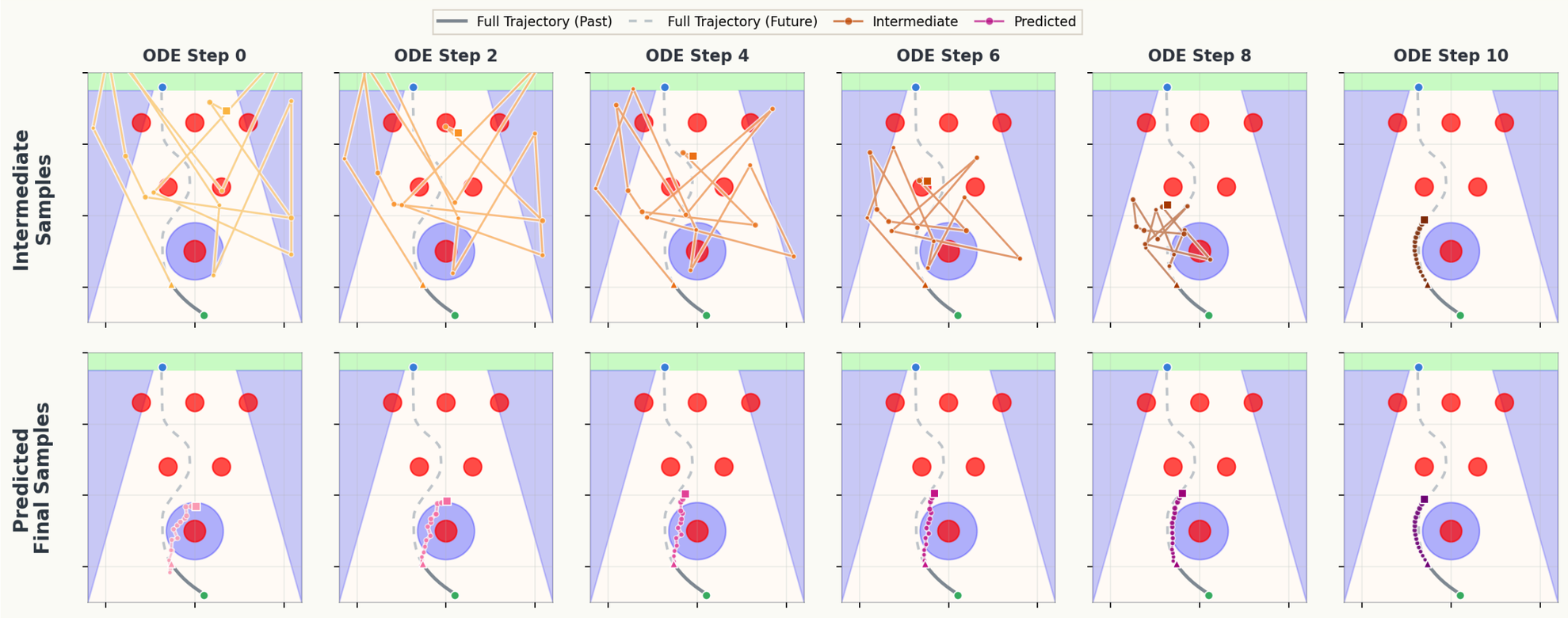
Endpoint projection (HardFlow): project the predicted final plan

iterate $\mathbf{x}_{\tau_k}$	→	model step $\bar{\mathbf{x}}_{k+1}$	→	predict the final plan $\bar{\mathbf{x}}_1$	→	**project the prediction** $\hat{\mathbf{x}}_1^\star = \Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_1)$	→	pull back damped by τ_{k+1}	→	next iterate $\mathbf{x}_{\tau_{k+1}}$

Optimal control. Steer the learned flow with the least control effort so that only the final sample is feasible: $\min_{\mathbf{u}} \lambda \int_0^1 \frac{1}{2} \|\mathbf{u}_\tau\|^2 \mathrm{d}\tau$ s.t. $\dot{\mathbf{x}}_\tau = v_\theta + \mathbf{u}_\tau$, $\mathbf{x}_1 \in \mathcal{Z}_f$. Solved one sampling step at a time, on the predicted final plan, so that the constraint set is not distorted by the network.

Both solve the same program on the same feasible set with the same solver, from a threshold step to the end of sampling. The terminal cost is zero here, so the HardFlow solve is the projection of the predicted plan.

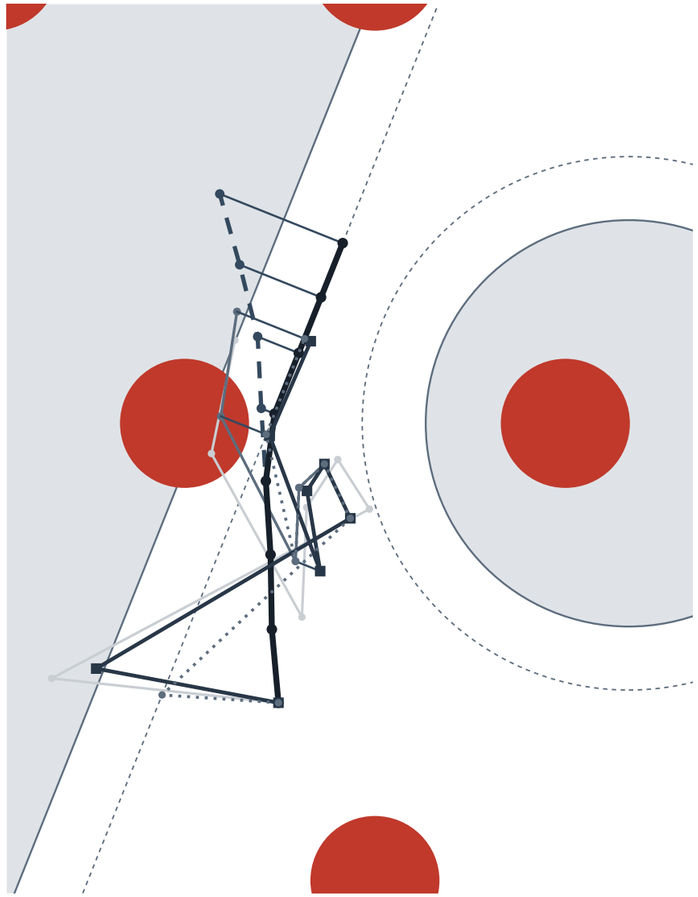
Endpoint projection constrains a trajectory from the first step on



- **Upper row:** the sample the solver holds, far from a trajectory until the last steps. Per-step projection poses its program on such an iterate.
- **Lower row:** its predicted final sample, a trajectory from the first step on: the point endpoint projection constrains.

Reproduced unaltered from Li, Alim and Azizan, HardFlow (arXiv:2511.08425v3), Fig. 11, CC BY 4.0; their task and run.

Endpoint projection, one guided step, on a plan



Illustration, not thesis data

The first guided step, $\tau_k = 0.5 \to \tau_{k+1} = 0.6$ of $K = 10$. The iterate is still noisy.

1 · Step the sampler once, unguided

$$\bar{\mathbf{x}}_{k+1} = \mathbf{x}_{\tau_k} + \Delta\tau\, v_\theta\big(\mathbf{x}_{\tau_k}, \tau_k, \mathbf{c}\big)$$

2 · Predict the final plan, in closed form

$$\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_{k+1} + \big(1 - \tau_{k+1}\big)\, v_\theta\big(\bar{\mathbf{x}}_{k+1}, \tau_{k+1}, \mathbf{c}\big)$$

3 · Project the predicted plan

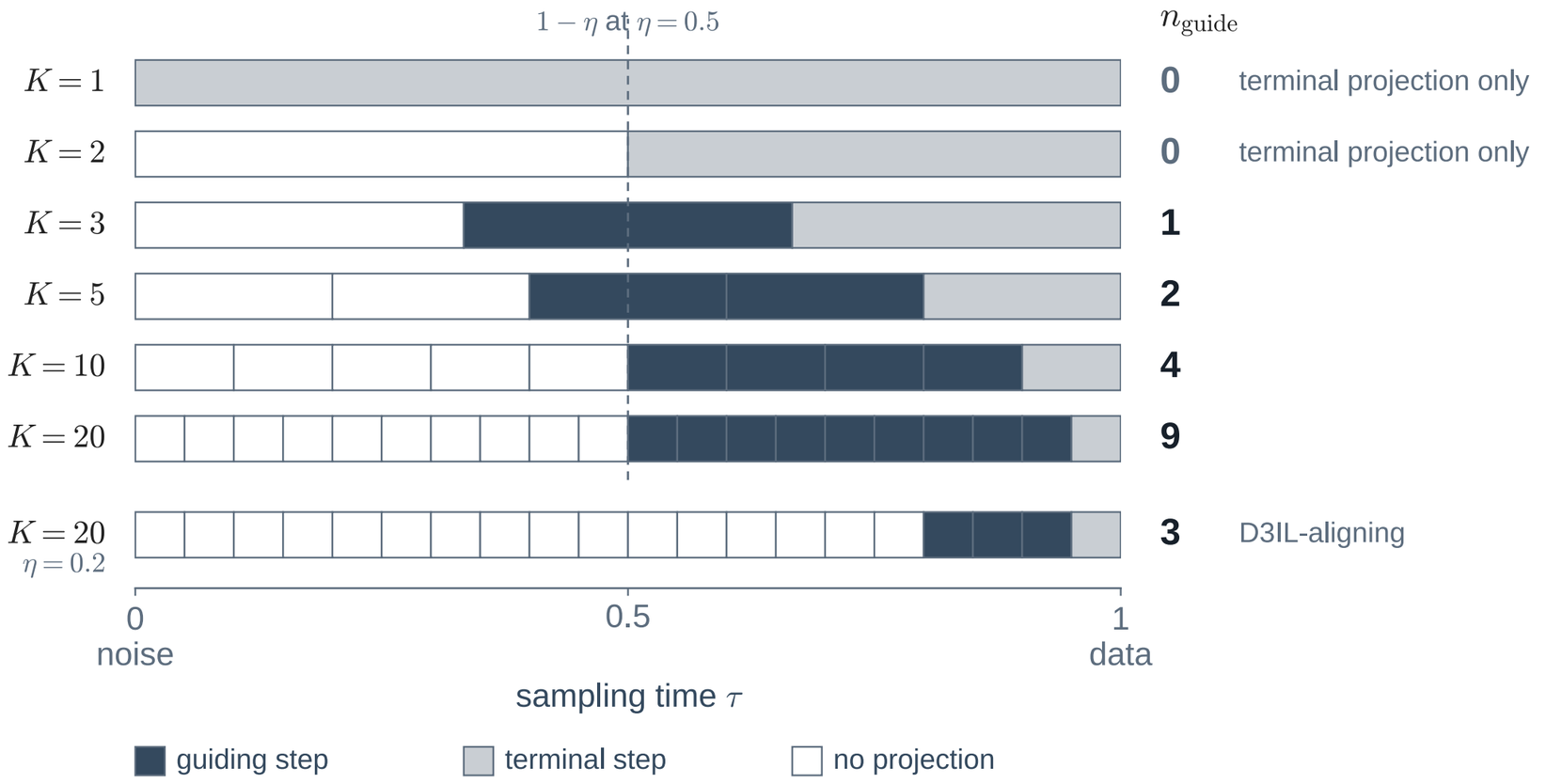
$$\hat{\mathbf{x}}_1^\star = \Pi_{\mathcal{Z}_f}\big(\bar{\mathbf{x}}_1\big)$$

4 · Pull back the correction, damped by τ_{k+1}

$$\mathbf{x}_{\tau_{k+1}} = \bar{\mathbf{x}}_{k+1} + \tau_{k+1}\big(\hat{\mathbf{x}}_1^\star - \bar{\mathbf{x}}_1\big)$$

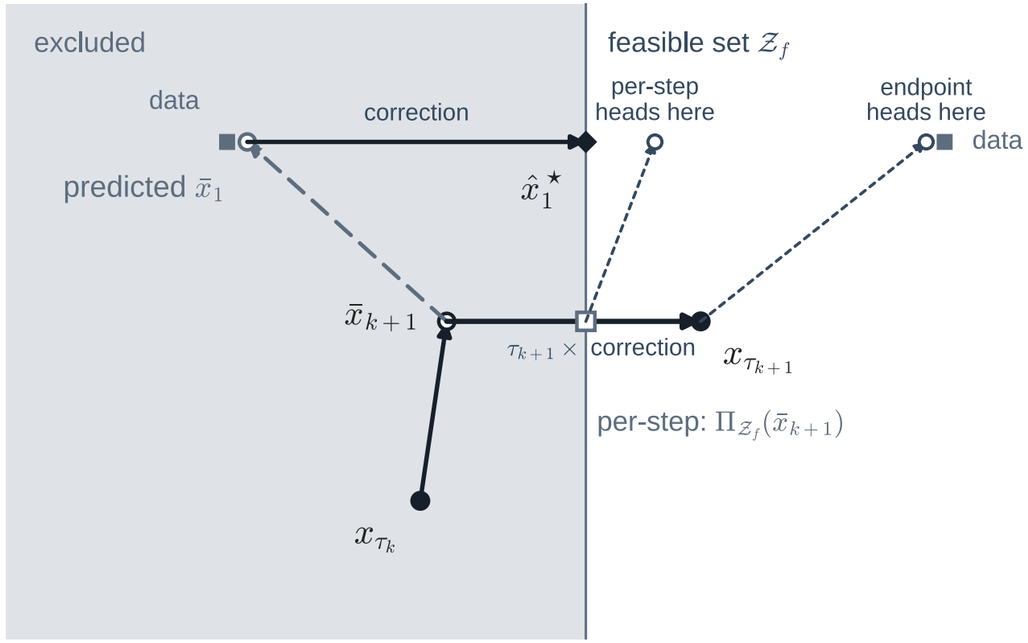
Per-step projection would project the noisy iterate itself, $\Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_{k+1})$.

Endpoint projection guides only where the budget leaves a step before the last



$n_{\text{guide}}(K, \eta) = \lceil \eta K \rceil - 1$ for $\eta > 0$: the active steps are the last $\lceil \eta K \rceil$, and the terminal one does not guide.

Per-step projection constrains the current iterate, endpoint projection a predicted final plan



Illustration, not thesis data

Each point stands for a whole plan: the squares are two demonstrated plans, the white half is the feasible set $\mathcal{Z}_f$. Projecting moves a point to the closest feasible point.

1 · Step the sampler once, unguided

$$\bar{\boldsymbol{x}}_{k+1} = \boldsymbol{x}_{\tau_k} + \Delta\tau\, v_\theta\big(\boldsymbol{x}_{\tau_k}, \tau_k, \boldsymbol{c}\big)$$

2 · Predict the final point: here, the excluded data point

$$\bar{\boldsymbol{x}}_1 = \bar{\boldsymbol{x}}_{k+1} + \big(1-\tau_{k+1}\big)\, v_\theta\big(\bar{\boldsymbol{x}}_{k+1}, \tau_{k+1}, \boldsymbol{c}\big)$$

3 · Project the prediction to the closest feasible point

$$\hat{\boldsymbol{x}}_1^\star = \arg\min_{\hat{\boldsymbol{x}}_1 \in \mathcal{Z}_f} \big\|\hat{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_1\big\|_2^2$$

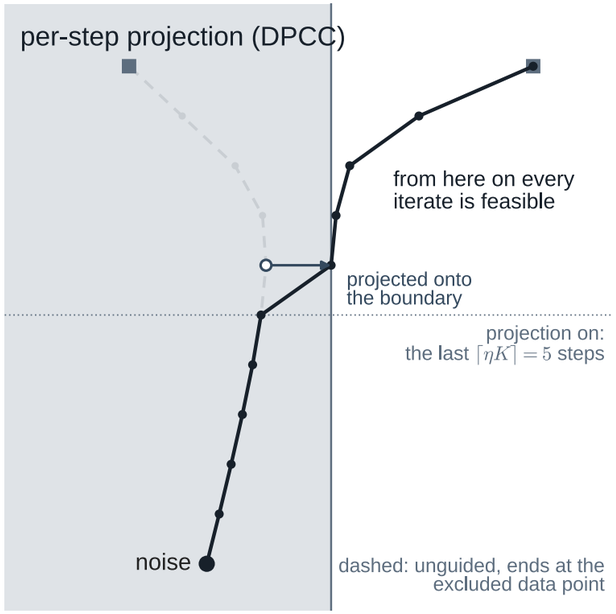
4 · Pull back: move the iterate by τ_{k+1} times the same correction

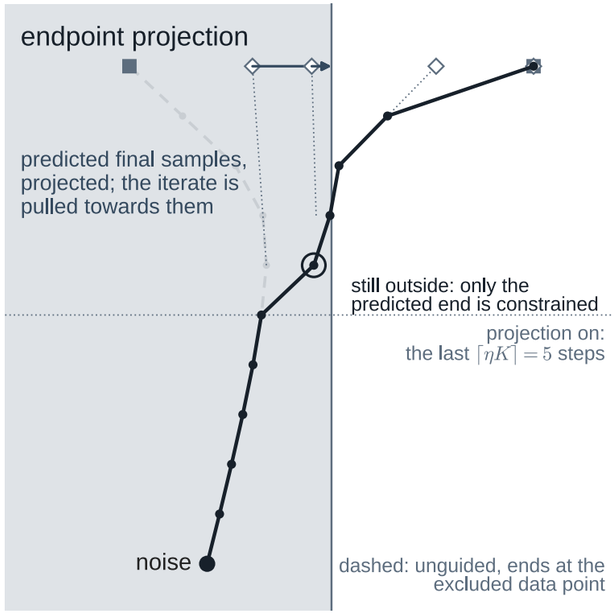
$$\boldsymbol{x}_{\tau_{k+1}} = \bar{\boldsymbol{x}}_{k+1} + \tau_{k+1}\big(\hat{\boldsymbol{x}}_1^\star - \bar{\boldsymbol{x}}_1\big)$$

- **Per-step (DPCC)** projects the iterate itself: $\boldsymbol{x}_{\tau_{k+1}} = \Pi_{\mathcal{Z}_f}\big(\bar{\boldsymbol{x}}_{k+1}\big)$.

Where each now heads: per-step, between the two data points; endpoint, to the feasible one.

One sampling run: per-step against endpoint projection



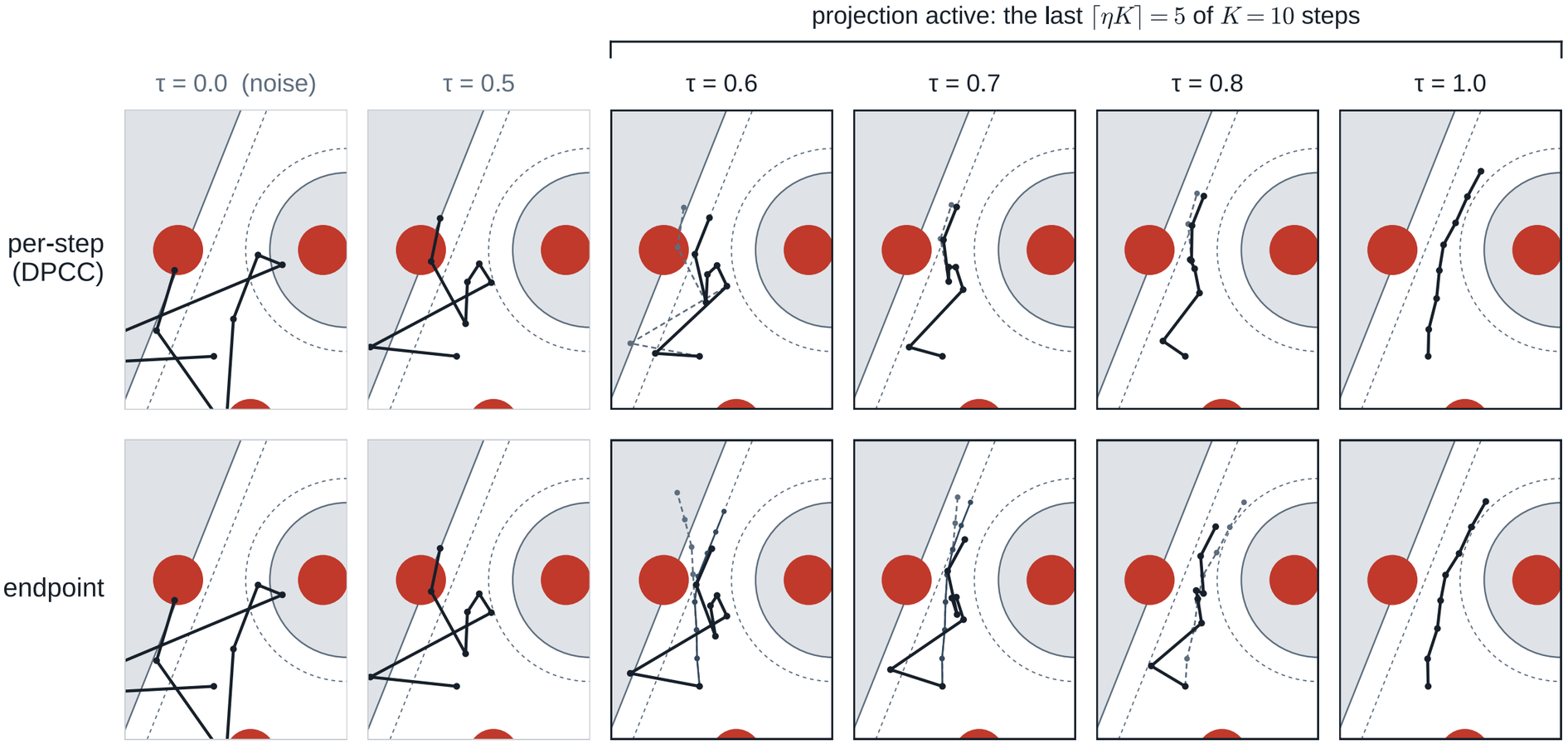


- **Per-step (DPCC)** corrects the current iterate. From the threshold on, every iterate is projected and therefore feasible; here only the first projection moves it.
- **Endpoint** corrects the plan the sampler is heading towards. The iterate follows, damped by τ, and can itself still be outside. $\mathrm{NFE} = K + n_{\mathrm{guide}} = 10 + 4$.

At the last step both project the finished sample: both end at the feasible data point, the unguided run at the excluded one.

Illustration, not thesis data

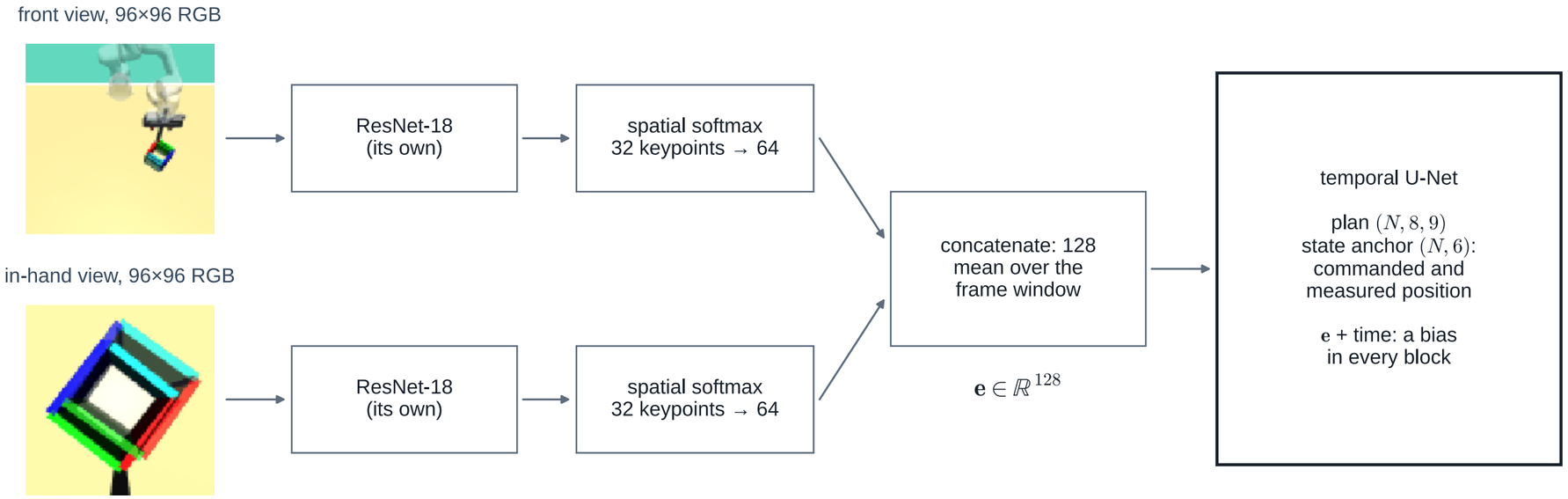
One sampling run, step by step: per-step against endpoint projection



Per-step: the Euler step (dashed) is projected (solid). **Endpoint:** the predicted plan (dashed) is projected (thin), and the iterate (solid) follows.

Illustration, not thesis data

D3IL-aligning: the images enter the network as one latent of 128 numbers



- The observation is two RGB views plus a 6-D state, the commanded and measured end-effector positions; the box and its target pose are only in the images.
- The backbone never sees an image: the latent $\mathbf{e}$ is the interface between perception and generation; the encoders train end to end with it.

Frames: thesis Fig. 5.5

The quadrotor keeps the planning side and changes the control problem

D3IL · manipulator

observation (p_{des}, p) → plan $(N, 8, d)$ → projection Euler rows → selection → setpoint $+\Delta p$ → **differential IK → joint PD**

UAV · quadrotor

observation $(p_{\text{des}}, p, \dot{p})$ → plan $(N, 8, d)$ → projection Euler rows → selection → setpoint $+\Delta p$ → **geometric cascade → rotor thrusts**

Stays: the planning side

- plans of eight position increments; the same setpoint recursion
- the projection with its first-order dynamics rows ($t_s = 1$)
- the four objectives and their training targets; UAV-pillars flies the D3IL-avoiding models unchanged

Changes: the control problem

- an underactuated, open-loop unstable plant: four rotor thrusts for six degrees of freedom
- the tracking controller: cascaded geometric (MuJoCo MPC in the controller comparison)
- constraints that leave the horizontal plane (UAV-corridor); generated demonstrations

The quadrotor controller: a cascaded geometric tracking law, PD at both levels, no integral term

Outer loop, position → thrust vector: a diagonal PD on position and velocity error with acceleration feed-forward; the gains are stiffer in the vertical channel. The commanded velocity is zero in every reported flight, so the position error drives the loop and the velocity term damps it.

$$\boldsymbol{a}_{\mathrm{cmd}} = -\mathbf{K}_p\,\boldsymbol{e}_p - \mathbf{K}_d\,\boldsymbol{e}_v + \ddot{\boldsymbol{p}}_{\mathrm{des}}, \qquad \boldsymbol{F} = m\,(\boldsymbol{a}_{\mathrm{cmd}} + g\,\boldsymbol{e}_3), \qquad \boldsymbol{e}_p = \boldsymbol{p} - \boldsymbol{p}_{\mathrm{des}},\ \ \boldsymbol{e}_v = \dot{\boldsymbol{p}} - \dot{\boldsymbol{p}}_{\mathrm{des}}$$

Attitude extraction: the rotor-disk normal points along $\boldsymbol{F}$; the commanded yaw fixes the remaining axis, giving $\mathbf{R}_{\mathrm{des}}$.

$$\boldsymbol{b}_{3,\mathrm{des}} = \frac{\boldsymbol{F}}{\|\boldsymbol{F}\|}, \qquad \boldsymbol{b}_{2,\mathrm{des}} = \frac{\boldsymbol{b}_{3,\mathrm{des}} \times \boldsymbol{x}_c}{\|\boldsymbol{b}_{3,\mathrm{des}} \times \boldsymbol{x}_c\|}, \qquad \boldsymbol{b}_{1,\mathrm{des}} = \boldsymbol{b}_{2,\mathrm{des}} \times \boldsymbol{b}_{3,\mathrm{des}}, \qquad \boldsymbol{x}_c = (\cos\psi_{\mathrm{des}}, \sin\psi_{\mathrm{des}}, 0)^\top$$

Inner loop, geometric attitude control: the coordinate-free attitude error on $SO(3)$ and a PD moment; the commanded angular velocity is zero, so $\boldsymbol{e}_\omega = \boldsymbol{\omega}$.

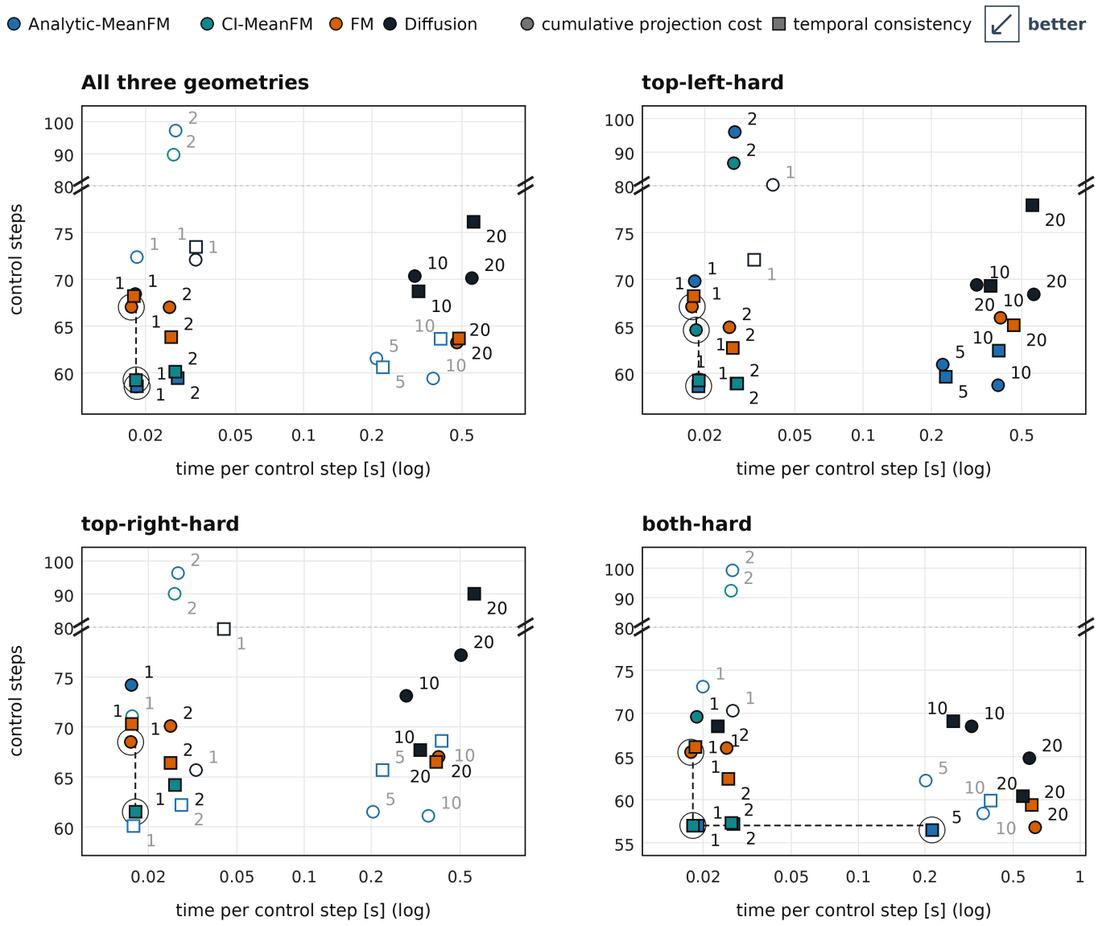
$$\boldsymbol{e}_R = \tfrac{1}{2}\left(\mathbf{R}_{\mathrm{des}}^\top \mathbf{R} - \mathbf{R}^\top \mathbf{R}_{\mathrm{des}}\right)^\vee, \qquad \boldsymbol{\tau}_b = -\mathbf{K}_R\,\boldsymbol{e}_R - \mathbf{K}_\omega\,\boldsymbol{e}_\omega + \boldsymbol{\omega} \times \mathbf{J}_b\,\boldsymbol{\omega}$$

Allocation: collective thrust = the world force on the body z axis, with a floor; the wrench is distributed over the four rotors by the geometry matrix, rescaled thrust-first and clipped to the actuator box.

$$T = \max\left(\boldsymbol{F}^\top \mathbf{R}\,\boldsymbol{e}_3,\ T_{\mathrm{floor}}\right), \qquad \mathbf{M}\,\boldsymbol{u} = \begin{bmatrix} T \\ \boldsymbol{\tau}_b \end{bmatrix}$$

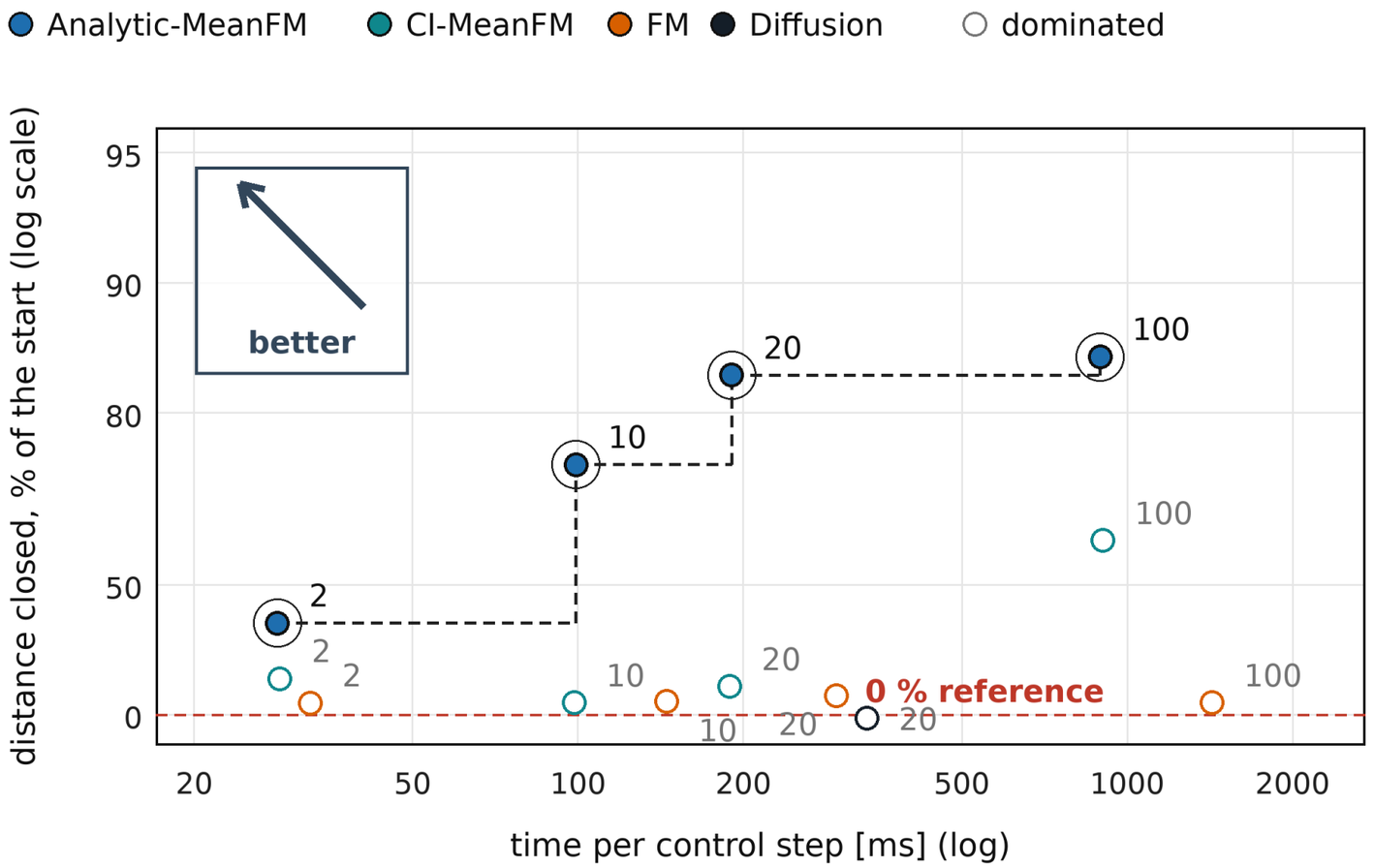
Parameters of record: $\mathbf{K}_p = \mathrm{diag}(4,4,8)$, $\mathbf{K}_d = \mathrm{diag}(3,3,4)$, $\mathbf{K}_R = \mathrm{diag}(70,70,4)$, $\mathbf{K}_\omega = \mathrm{diag}(2.5,2.5,1)$; $u_{\mathrm{hover}} = mg/4$, $u_{\max} = \max(2u_{\mathrm{hover}}, 6)$, $u_{\min} = 0$, $T_{\mathrm{floor}} = 0.1\,mg$; m and $\mathbf{J}_b$ from the vehicle model. The controller runs at every physics step (10 ms); the plan is queried every third one (30 ms; every hundredth, 1 s, on UAV-pillars). **It is tagged `pid` in the run folders, though the loop has no integral term.**

Thesis figure: D3IL-avoiding, every budget and rule



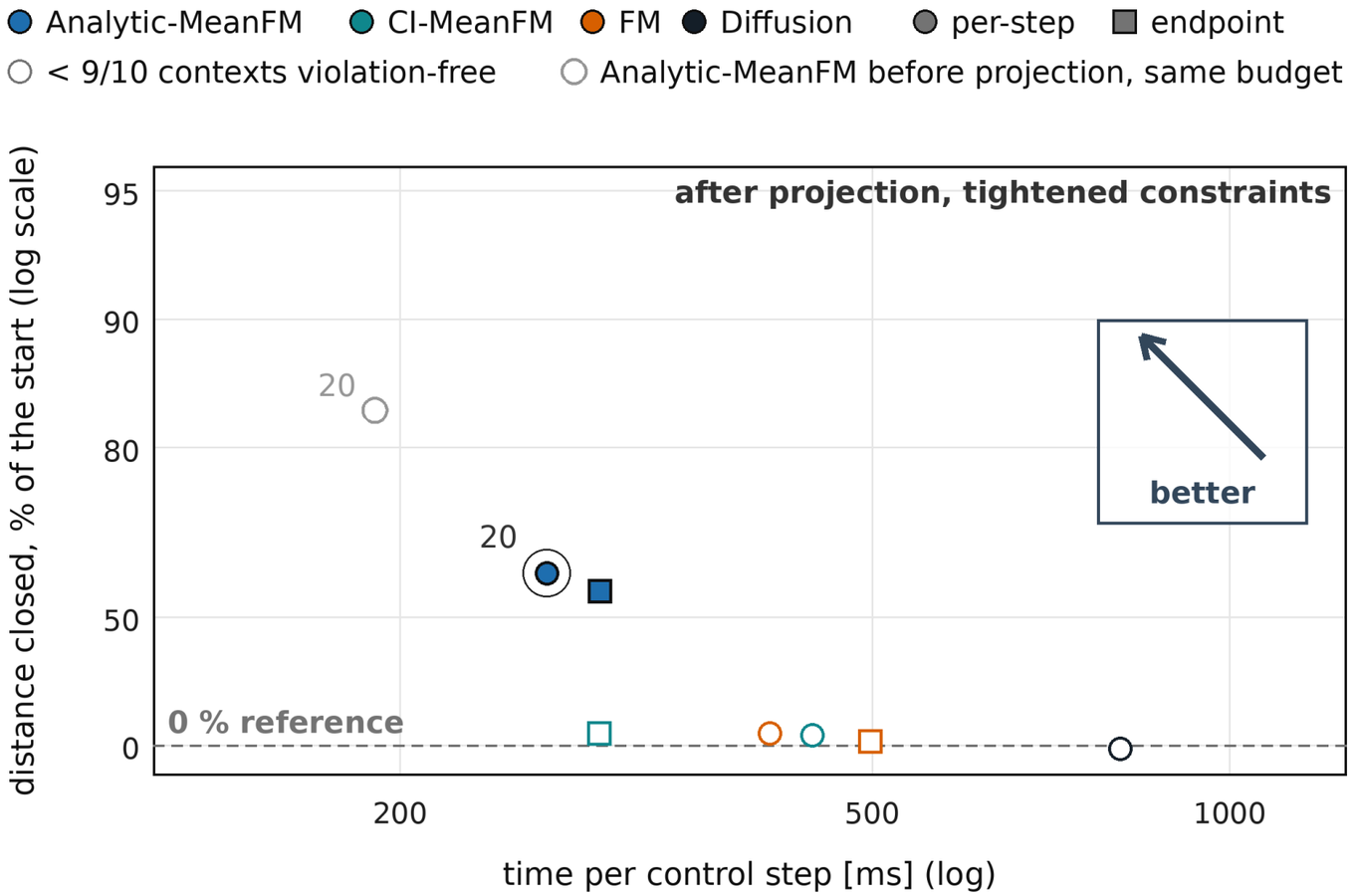
Control steps against time per control step, every budget and selection rule, all three geometries. Thesis Fig. 6.2

Thesis figure: D3IL-aligning, distance closed against time



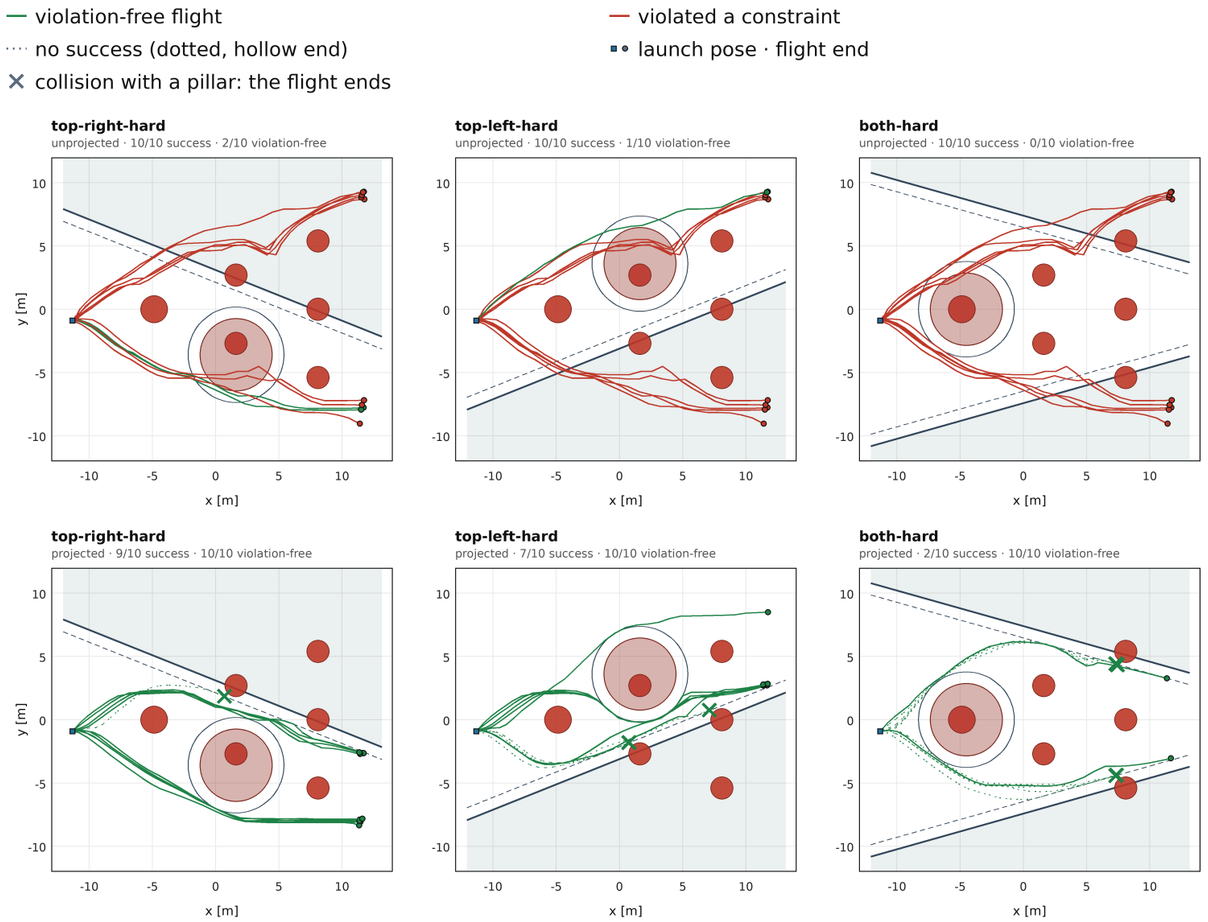
Distance closed against time per control step, every budget, without projection. Thesis Fig. 6.3

Thesis figure: D3IL-aligning under projection



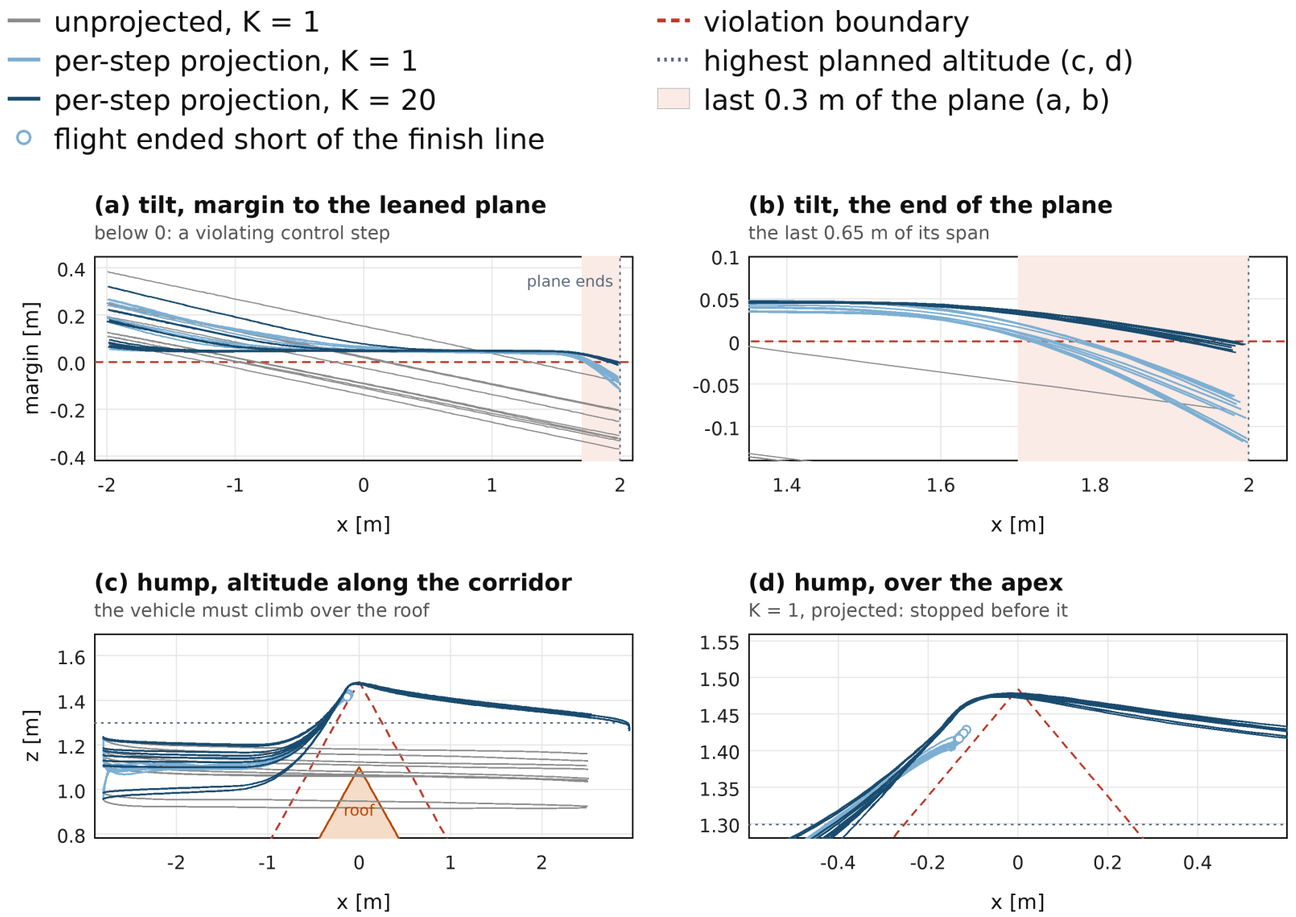
Distance closed against time per control step after projection; circles per-step, squares endpoint; filled points keep at least nine of ten contexts free of violations. Thesis Fig. 6.4

Thesis figure: UAV-pillars, the flown paths



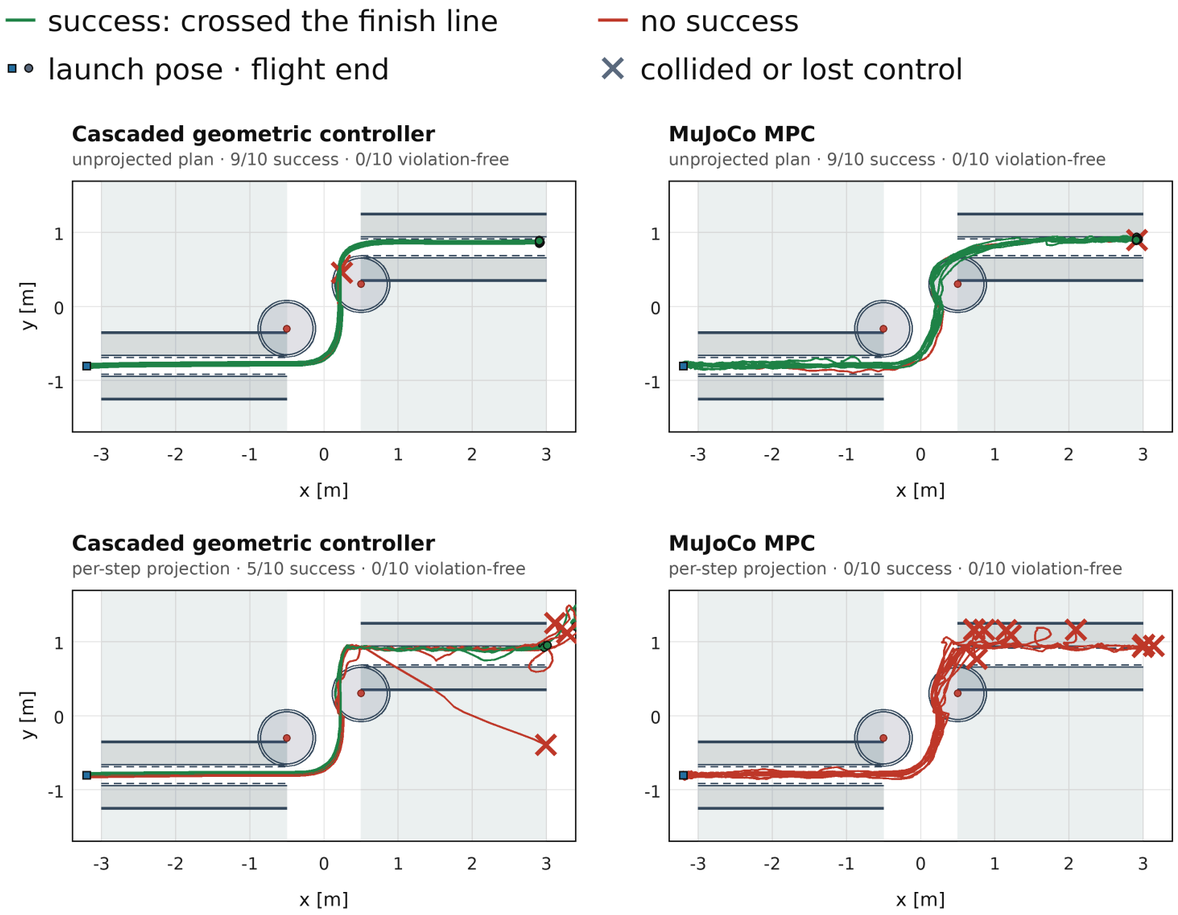
Flown paths, unprojected (top) and projected (bottom), three geometries. Thesis Fig. 6.5

Thesis figure: UAV-corridor, margin and altitude along the corridor



Margin to the tilt and altitude over the hump along the corridor: unprojected, per-step projection at K = 1 and K = 20; panel (d): the K = 1 flight stops before the apex. Thesis Fig. B.4

Thesis figure: UAV-s-curve, the flown paths



Cascaded geometric controller (left) and MuJoCo MPC (right); unprojected (top) and per-step projection (bottom). Thesis Fig. 6.7